

Comparison of MPC, Task-Space PID and Time-delayed Control

on a 3-DOF OpenManipulator-X

Stage Report

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Abstract

This stage report compares three controllers holding a task-space reference on the same 3-DOF waist-removed OpenManipulator-X: a task-space MPC with a Kalman disturbance observer, a task-space PID baseline and joint-space Time-delayed Control (TDC). All three are driven by the identical step reference, at the identical 100 Hz loop rate, scored with the identical metrics, at two different arm postures, with every headline number taken from three runs of eight steps ($n = 24$).

The single sentence that organises the results is that **this arm is friction-dominated, so what separates the three controllers is how much authority each can bring to bear against friction and what limits that authority**. The PID draws its authority from K_p^{PID} and K_i^{PID} , both bounded by a loop-delay wall; at the second posture that wall forces a 73% gain cut and the arm stalls 4.6 mm short of the setpoint. The MPC meets the same wall but has a second channel in its observer term, which supplies the majority of its steady-state authority and therefore survives the gain cut almost unchanged. TDC’s authority is set by its nominal inertia $\bar{D}$, which does not lie on that wall at all: retuning $\bar{D}$ — the method’s own free parameter, with the control law left exactly as published — gave it enough authority to break stiction quickly at both postures.

Measured settled error, $n = 24$ per cell: TDC 0.395 and 0.428 mm at the two postures, MPC 0.504 and 0.864 mm, PID 0.663 and 2.823 mm. TDC is the only one of the three whose settling time is unchanged between the two postures (1.185 against 1.192 s) and it is also the cheapest per tick (2.7 ms mean against the MPC’s 8.2 ms, whose 99th percentile already exceeds the 10 ms control period).

The clearest single piece of evidence for the friction reading is a negative one: the TDC configuration that produced the best result of the whole study on hardware *diverges within 0.45 s in simulation*, because the simulation contains no Coulomb friction and the authority that $\bar{D}$ buys on hardware is, there, merely excess loop gain.

TDC acts in joint space and its input is three joint angles, so the three controllers cannot be placed on one common gain parameterisation; what is and is not equalised by the comparison is stated in Section 4.5.

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1 Introduction and Objectives

The target application is physical human–robot interaction (pHRI), where a manipulator must hold a task-space reference accurately while remaining compliant enough to be safe on contact. This stage of the project evaluates three controllers against that goal on one arm:

- a **task-space MPC with a Kalman disturbance observer**, following Cao & Tang (arXiv:2606.08281), in the receding-horizon QP realization of their double-integrator MPC;
- a **task-space PID**, the baseline the other two have to beat;
- **Time-delayed Control (TDC)** in its basic joint-space form, following Memmer’s thesis (Ch. 3.2.1 / 4.1 / 4.2).

The objectives for this stage were: (i) bring all three controllers to a verified, repeatable operating point on real hardware; (ii) establish *why* each one ends up where it does, rather than merely reporting its score; and (iii) compare them under conditions in which the reference trajectory, the loop rate and the evaluation metrics are the same for all three.

What the report argues. A ranking on its own would not be worth much, because the three controllers are not interchangeable objects — TDC in particular acts in joint space, which Section 4.5 treats explicitly. The claim the report actually makes is about a mechanism:

On this arm the binding constraint is *friction authority*. Every result below — the stability boundary, the asymmetry between the two postures, the observer’s contribution, the effect of retuning $\bar{D}$ and the fact that the simulation inverts the hardware verdict — is an instance of how much torque a controller can bring to bear against stiction and of what stops it bringing more.

Section 2.4 already shows the arm from that angle before any controller is discussed: the rigid-body model of Section 2.2 contains no friction term at all and consequently mispredicts the arm’s motion by two orders of magnitude on one joint.

The report is a progress report. It presents controller performance and the staged results behind it; it does not claim a general ranking of the three methods beyond this arm and this task.

2 Hardware and the Dynamics Model

2.1 The manipulator

The platform is a ROBOTIS OpenManipulator-X with the waist joint physically removed, leaving three actuated joints driven by XM430-W350 servos with IDs 12, 13 and 14. All three joints pitch about the same axis, so the end-effector moves in a single vertical plane and the task space used throughout this report is the two-dimensional (x, z) plane.

All servos run in **Current Control Mode**: the controller computes a joint torque every tick and writes it to the `Goal_Current` register. No position or velocity loop inside the servo is active, so the control law under test is the only thing closing the loop. The control period is **10 ms (100 Hz)** for every result in this report.

Figure 1 shows the kinematic chain with its measured link offsets and the two arm postures the report uses. The chain is a fixed base offset followed by three pitch joints,

$$\ell_0 = (12, 0, 59.5), \quad \ell_1 = (24, 0, 128), \quad \ell_2 = (124, 0, 0), \quad \ell_3 = (126, 0, 0) \text{ mm},$$

so that with $R_y(\cdot)$ a rotation about the pitch axis the joint origins and the tool point follow from

$$\mathbf{p}_{k+1} = \mathbf{p}_k + R_y(q_1) \cdots R_y(q_k) \mathbf{d}_k, \quad k = 0, 1, 2, 3. \quad (1)$$

The task Jacobian $J_{xz}(\mathbf{q}) \in \mathbb{R}^{2 \times 3}$ is the two rows of $\partial \mathbf{p}_4 / \partial \mathbf{q}$ corresponding to x and z . Three joints serve a two-dimensional task, so the arm has one degree of redundancy, which the three controllers use differently (Section 4.5).

The 3-DOF waist-removed OpenManipulator-X

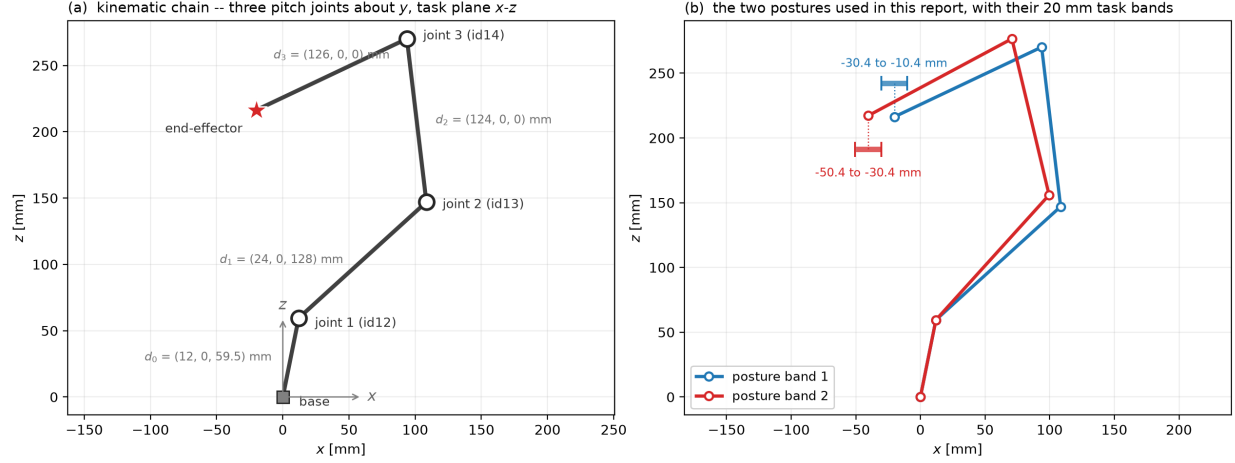


Figure 1: The 3-DOF waist-removed OpenManipulator-X. (a) the kinematic chain with its link offsets; (b) the two postures used in this report, with the 20 mm task band each of them explores.

2.2 Rigid-body dynamics

For a torque-controlled serial manipulator the equations of motion are

$$M(\mathbf{q})\ddot{\mathbf{q}} + C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) = \boldsymbol{\tau} + J(\mathbf{q})^\top \mathbf{F}_h, \quad (2)$$

with $M(\mathbf{q})$ the positive-definite joint-space inertia matrix, $C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}}$ the Coriolis and centrifugal vector, $\mathbf{G}(\mathbf{q})$ the gravity torque and $\mathbf{F}_h$ any external contact wrench — unmeasured on this arm, since it carries no force sensor. Because all three joints pitch about a common axis, every rotation composes by simple angle addition and expanding the Lagrangian kinetic and potential energy for this geometry gives closed forms for all three terms:

$$M(\mathbf{q}) = \sum_{i=1}^3 \left[m_i J_{v,ci}(\mathbf{q})^\top J_{v,ci}(\mathbf{q}) + J_{\omega,i}(\mathbf{q})^\top R_{0i}(\mathbf{q}) I_i R_{0i}(\mathbf{q})^\top J_{\omega,i}(\mathbf{q}) \right], \quad (3)$$

$$[\mathbf{G}(\mathbf{q})]_k = g \sum_{i=1}^3 m_i [J_{v,ci}(\mathbf{q})]_{z,k}, \quad g = 9.81 \text{ m/s}^2, \quad (4)$$

$$[C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}}]_k = \sum_{i=1}^3 \sum_{j=1}^3 \Gamma_{ijk}(\mathbf{q}) \dot{q}_i \dot{q}_j, \quad \Gamma_{ijk} = \frac{1}{2} \left(\frac{\partial M_{kj}}{\partial q_i} + \frac{\partial M_{ki}}{\partial q_j} - \frac{\partial M_{ij}}{\partial q_k} \right), \quad (5)$$

where $J_{v,ci}$ and $J_{\omega,i}$ are the per-link translational and rotational Jacobians and Γ_{ijk} are the Christoffel symbols of $M(\mathbf{q})$ — the choice that keeps $\dot{M} - 2C$ skew-symmetric.

These expressions are not evaluated symbolically in the implementation. A single recursive Newton–Euler routine is called three times per tick at chosen evaluation points: $\dot{\mathbf{q}} = \mathbf{0}$, $\ddot{\mathbf{q}} = \mathbf{0}$ returns $\mathbf{G}(\mathbf{q})$; $\ddot{\mathbf{q}} = \mathbf{0}$ with gravity switched off returns $C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}}$; and unit acceleration vectors return the columns of $M(\mathbf{q})$. This is exact rather than approximate, because recursive Newton–Euler evaluates the left-hand side of Eq. (2) exactly for any $(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}})$.

Table 1: Link inertial parameters, taken from the manufacturer’s URDF with the link indices renumbered after the waist joint’s physical removal. $\mathbf{c}_i$ is the centre of mass in the link’s own frame.

link i	m_i [kg]	$\mathbf{c}_i$ [m]	offset from parent [m]
1	0.13851	(0.01031, 0.00038, 0.10170)	(0.012, 0, 0.0595)
2	0.13275	(0.09091, 0.00039, 0.00022)	(0.024, 0, 0.128)
3	0.14328	(0.04421, 0, 0.00891)	(0.124, 0, 0)

Link and joint are kept distinct. “Link i ” is the rigid body of Table 1; “joint i ” is the actuator (servo id12, id13, id14 for $i = 1, 2, 3$) that rotates link i relative to link $i - 1$ and is what a commanded torque and a friction force act on. Joint 3 drives link 3, which has the smallest rotational inertia of the three about its own dominant axis ($I_{3,yy} = 7.6 \cdot 10^{-5}$ against $3.4 \cdot 10^{-4}$ and $2.4 \cdot 10^{-4}$ kg m² for links 1 and 2). Since link 3 is the only body joint 3 has to accelerate, that is the inertia resisting its own dynamics — a fact that decides the outcome of Section 2.5.

2.3 Task-space quantities

Two quantities derived from $M(\mathbf{q})$ and $J_{xz}(\mathbf{q})$ are used repeatedly later — in the QP’s own plant model (Section 3.1), in the gain mapping of Section 4.1 and in the stability criterion of Section 4.2 — so both are defined here.

Task-space inertia. The first is

$$\mathbf{\Lambda}(\mathbf{q}) = \left(J_{xz} M_{\text{used}}^{-1} J_{xz}^T + \mu I_{2 \times 2} \right)^{-1}, \quad \mu = 0.05, \quad (6)$$

with M_{used} the corrected inertia matrix of Eq. (9) below.

$\mathbf{\Lambda}$ has a direct physical reading and it is worth stating plainly because the whole of Section 4.1 rests on it. From Eq. (11), a task-space force applied at the tool produces $\ddot{\mathbf{p}} = \mathbf{\Lambda}^{-1} \mathbf{F}$: **$\mathbf{\Lambda}$ is the apparent mass the end-effector presents when it is pushed**, in kilograms. At the postures used in this report $\Lambda_{xx} \approx 0.67$ kg and $\Lambda_{zz} \approx 0.43$ kg, so pushing the tool horizontally feels about 1.6× heavier than pushing it vertically. Three properties follow:

- It is a *matrix*, not a scalar, because the apparent mass depends on direction. The off-diagonal entry is the coupling: a force along x also produces acceleration along z .
- It is *pose-dependent*, because it is the joint-space inertia projected to the tool through the Jacobian. Both $M(\mathbf{q})$ and $J_{xz}(\mathbf{q})$ change with configuration, so an extended arm presents more apparent mass at the tool than a folded one.

- It is what makes “the gain” a pose-dependent quantity in Section 4.1: the same LQR weights map to different task-space stiffness at different postures.

The damping term μ keeps the inverse in Eq. (6) conditioned near the workspace boundary, where the bracketed matrix becomes nearly singular so that Λ would otherwise grow without bound. It caps the apparent mass at $1/\mu = 20$ kg. It is not free: at the first posture it reduces Λ_{xx} from 0.6963 to 0.6729 kg, a bias of 3.4 %, which is the price paid for that conditioning.

Null-space projector. The second is

$$\mathcal{N}(\mathbf{q}) = I_3 - J_{xz}^\top (\Lambda J_{xz} M_{\text{used}}^{-1}), \quad (7)$$

which has the property $J_{xz} M_{\text{used}}^{-1} \mathcal{N}(\mathbf{q}) = 0$ identically. Any joint torque of the form $\mathcal{N}(\mathbf{q}) \boldsymbol{\tau}_{\text{post}}$ therefore produces *no* task-space acceleration at all: it can be used to steer the arm’s redundant degree of freedom (Section 2.1) into a preferred posture without disturbing the task. This identity is the step that removes the posture term from the feedback-linearisation derivation in Section 3.1.

2.4 Validating the model against hardware

The model above is only useful to the extent that it matches the real arm. Two open-loop tests were run, both with the feedback loop deliberately absent, so that what is measured is the model’s own accuracy rather than a controller’s ability to paper over its errors.

Static: gravity alone. With the arm feedback-free and commanded $\boldsymbol{\tau} = \mathbf{G}(\mathbf{q}) - b_{\text{safe}} \dot{\mathbf{q}}$ ($b_{\text{safe}} = 0.15$ Nm/(rad/s), pure safety damping, no position feedback of any kind), the joints were left to hold themselves. Over the settled portion of a 15 s run ($t \geq 4.6$ s, 682 samples) the joint angles are motionless to

$$\sigma_{\mathbf{q}} = (5.4 \cdot 10^{-4}, 2.0 \cdot 10^{-4}, 8.3 \cdot 10^{-5}) \text{ rad} \approx (0.031^\circ, 0.012^\circ, 0.0048^\circ), \quad (8)$$

which is within encoder and servo-loop noise (Figure 2). This rules out a systematic error in the gravity parameters of Table 1 — *including for joint 3*, whose static holding at this pose is the cleanest of the three. Whatever is wrong with the model only appears once the arm moves.

Dynamic: replaying the measured torque. Each joint was then jogged individually with a torque sine (amplitude 0.35 Nm, period 4 s) and the logged commanded torque was replayed through Eq. (2).

Note on the sampling rate. These probe logs were recorded at $\Delta t = 10$ ms, the same tick as the controller’s own, so “one step” below is a genuine one-tick prediction rather than a coarser proxy for it. The probe itself is still a separate, feedback-free open-loop tool from the closed-loop controller — the matching rate is convenient, not a claim that the two are the same test.

The comparison uses n -step-ahead prediction rather than one continuous open-loop simulation: at every sample the model is reset to the real measured state, integrated forward n samples on the logged torque alone and compared against the state actually measured n samples later. A single continuous replay is not usable here, because the logged torque is not a pure function of time — it already contains gravity feed-forward and damping terms computed from the real measured state, so as soon as the prediction drifts the replayed torque is no longer the torque that would have been commanded and for a gravity-loaded arm that compounds without bound.

Static validation of the gravity model: all three joints hold within encoder noise

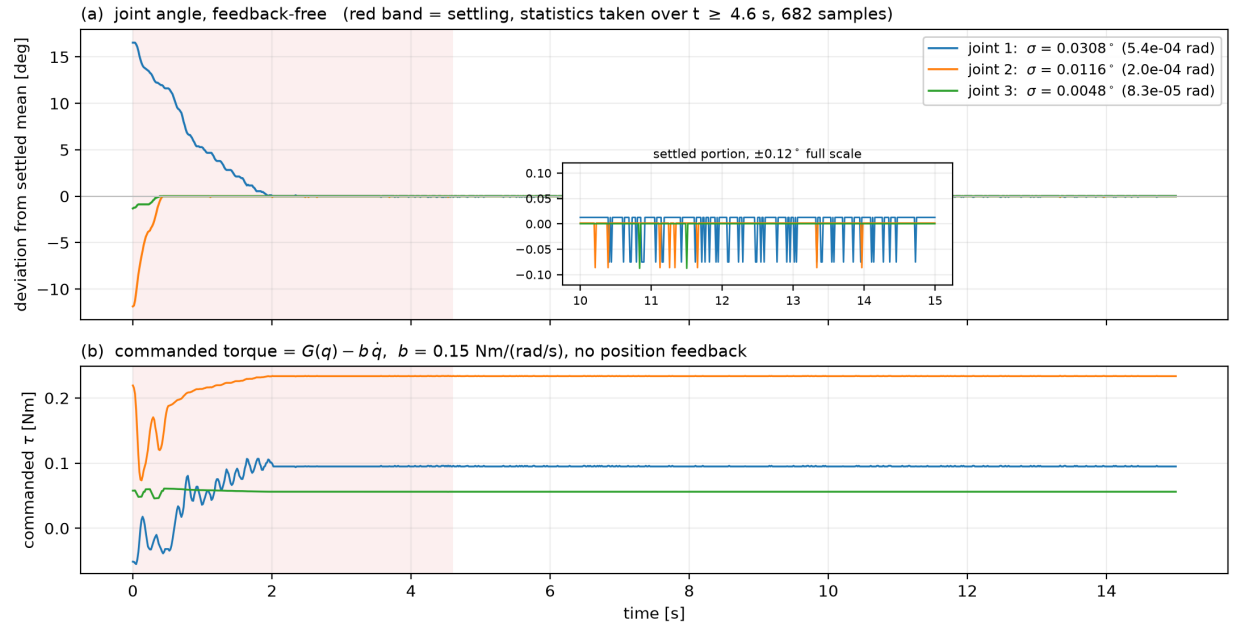


Figure 2: Static validation of the gravity model. The inset shows the settled portion at $\pm 0.12^\circ$ full scale.

Open-loop validation of the model as tabulated: the measured torque replayed through it, at two prediction horizons

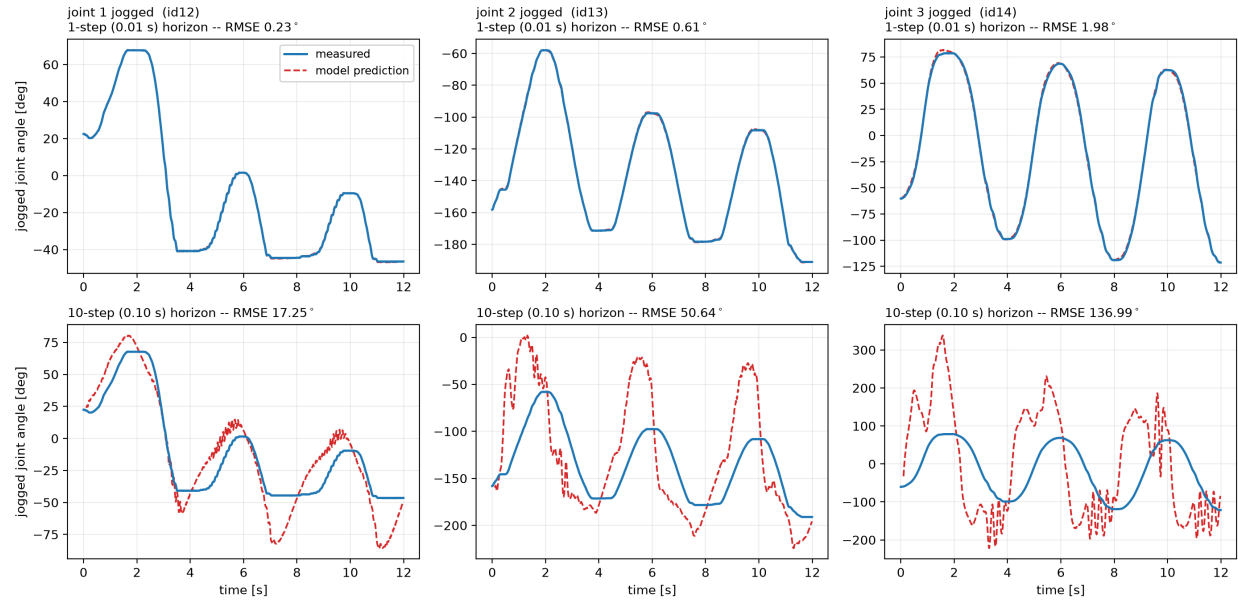


Figure 3: Open-loop validation of the model as tabulated, at two prediction horizons. Top row: one step. Bottom row: 0.1 s.

Table 2: Per-joint RMSE [deg] from the open-loop replay, model as tabulated. The jogged joint’s own entry is in bold. One step is one 10 ms sample of the probe log, matching the control loop’s own tick.

jogged joint	1-step (0.01 s) horizon			0.1 s horizon		
	joint 1	joint 2	joint 3	joint 1	joint 2	joint 3
1 (id12)	0.23	0.39	0.54	17.25	31.43	36.05
2 (id13)	0.25	0.61	0.82	20.50	50.64	54.03
3 (id14)	0.20	0.62	1.98	10.72	33.89	136.99

Two horizons are reported because they answer different questions (Table 2). At 0.1 s *every* joint diverges, so that horizon only says that the model is not good enough to predict 0.1 s ahead on a gravity-loaded arm with no friction term. The discriminating measurement is the one-step horizon, where joints 1 and 2 stay near half a degree (0.23°, 0.61°) while joint 3 does not (1.98°, three to nine times worse). The failure is therefore not diffuse; it is concentrated on one joint and it is the joint that Section 2.2 identified as having the least inertia of its own to resist an unmodelled torque.

2.5 Correcting the inertia model

The correction adopted is a constant added to the diagonal of the inertia matrix,

$$M_{\text{used}}(\mathbf{q}) = M(\mathbf{q}) + c I_{3 \times 3}, \quad c = 0.01 \text{ kg m}^2, \quad (9)$$

and this is what every controller in this report actually uses. The physical reading is that the parameter table of Section 2.2 is derived from link geometry and therefore contains no reflected rotor or gear-train inertia; a constant on the diagonal is the standard way of representing that missing term and it is not pose-dependent because the rotor and gear train are not.

The effect on the open-loop score is decisive. Replaying the same log through Eq. (2) with and without the constant, the 0.1 s prediction RMSE of joint 3 falls from 137.0° to 2.87° (Figure 4(a)).

Why 0.01 and what this test cannot decide. Sweeping the constant over two decades (Figure 4(b)) gives

c [kg m ²]	0	0.002	0.005	0.01	0.02	0.03	0.05	0.10
joint 3 RMSE [deg]	137.0	11.53	4.96	2.87	2.14	2.02	1.99	2.01

The curve drops by more than an order of magnitude (47.7×) between 0 and 0.01 and is then nearly *flat*: over the whole decade from 0.01 to 0.10 the score varies by a factor of 1.44, with a minimum so shallow (1.99° at 0.05 against 2.87° at 0.01) that it carries no information. The honest statement is therefore twofold. The open-loop test proves that *some* added diagonal inertia is required and sizes its order of magnitude; it *cannot identify the value*. The value actually used, $c = 0.01$, was selected on closed-loop grounds — it is the largest correction under which the gain ladder of Section 6.1 remained stable — not because it minimises this open-loop score, which it does not.

Correcting the inertia model: a constant added to the diagonal of $M(q)$ (log jogs joint 3; joints 1 and 2 are held)

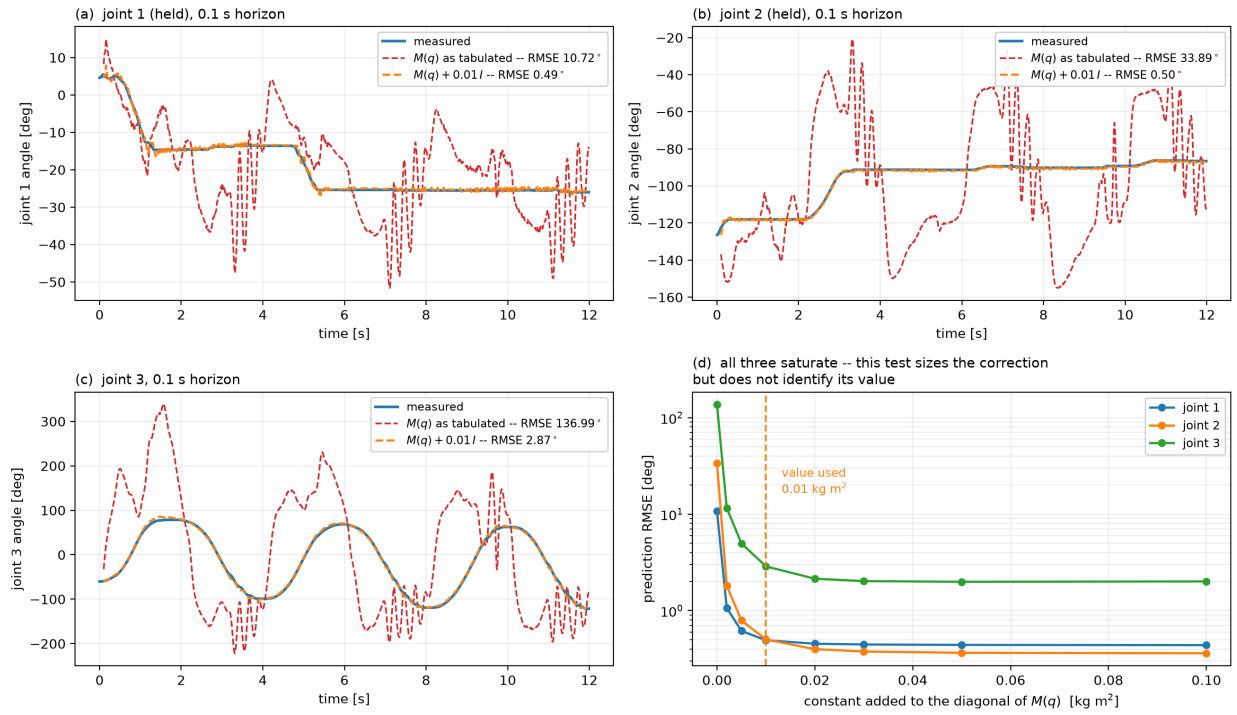


Figure 4: The inertia correction. (a) joint 3 predicted against measured, with and without the constant; (b) the same score swept over the value of the constant.

3 Controllers

Three controllers are described below and compared throughout. Because they are tuned in three different currencies and because two of them have gains that look alike but mean different things, the notation is fixed here and used without exception for the rest of the report.

Table 3: Notation. The letter says which space a quantity lives in, the superscript says which controller it belongs to. Undecorated K_p , K_d denote the task-space stiffness/damping *pair* as a physical property of an operating point — the axes of Figure 12 and the arguments of Eq. (22) — not any one controller’s setting.

	MPC	Task-space PID	TDC
acts in	task space (x, z)	task space (x, z)	joint space
input	task-space error $\mathbf{e}$	task-space error $\mathbf{e}$	joint error $\mathbf{e}_q$
output	task force $\mathbf{F}$	task force $\mathbf{F}$	joint torque $\boldsymbol{\tau}$
tuned by	q_{pos}, q_{vel}, r (dimensionless weights)	K_p^{PID} [N/m], K_i^{PID} , K_d^{PID} [Ns/m]	$\bar{D}$ [kg m ²], $K_p^{\text{TDC}}, K_d^{\text{TDC}}$
task-space gain	$K_p^{\text{MPC}}, K_d^{\text{MPC}}$ <i>derived</i> (Sec. 4.1)	$K_p^{\text{PID}}, K_d^{\text{PID}}$ <i>set directly</i>	none (Sec. 4.5)
limits	$\mathbf{F}_{\max}^{\text{MPC}} = 300$ N $\boldsymbol{\tau}_{\max}$ per joint	$F_{\max}^{\text{PID}} = 3.0$ N, $I_{\max}^{\text{PID}} = 0.25$	joint torque limit
extra state	$\hat{\mathbf{d}}$ from the observer, weights $q_{\hat{\mathbf{d}}}, r_y$	—	$\boldsymbol{\tau}(t - \lambda), \ddot{\mathbf{q}}(t - \lambda)$ delay $\lambda = \text{one tick}$

Symbols used elsewhere and never reused: $g = 9.81$ m/s²; $\ell_0 \dots \ell_3$ the link offsets of Section 2.1; $\mathbf{d}(t)$ the task-space disturbance of Eq. (11); $\mu = 0.05$ the regularisation in Eq. (6); $N = 10$ the QP horizon and $\mathcal{N}(\mathbf{q})$ the null-space projector; b the derivative setpoint weight of Eq. (23), which is 1.0 unless stated.

3.1 MPC with a Kalman disturbance observer

Feedback linearisation. The commanded joint torque is decomposed into a task channel and a base channel,

$$\boldsymbol{\tau} = \underbrace{J_{xz}(\mathbf{q})^\top \mathbf{F}_{task}}_{\text{task channel}} + \underbrace{\mathcal{N}(\mathbf{q})\boldsymbol{\tau}_{post} + \mathbf{G}(\mathbf{q}) + C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}}}_{\boldsymbol{\tau}_{base}}, \quad (10)$$

where $\boldsymbol{\tau}_{post}$ is a posture term acting only inside the null space of the task. Substituting Eq. (10) into the true dynamics Eq. (2) cancels $\mathbf{G}$ and $C\dot{\mathbf{q}}$ (they appear on both sides, exactly when the modelled and true values agree) and leaves $M\ddot{\mathbf{q}} = J_{xz}^\top \mathbf{F}_{task} + \mathcal{N}\boldsymbol{\tau}_{post}$. Left-multiplying by $J_{xz}M^{-1}$, the null-space term vanishes identically — $J_{xz}M^{-1}\mathcal{N} = 0$ is an algebraic identity of Eq. (7), not an approximation — and $J_{xz}M^{-1}J_{xz}^\top = \boldsymbol{\Lambda}^{-1}$ cancels the remaining $\boldsymbol{\Lambda}$, so that with $\mathbf{e} = \mathbf{p}_{xz} - \mathbf{p}_{d,xz}$,

$$\ddot{\mathbf{e}} = \boldsymbol{\Lambda}^{-1}(\mathbf{q}) \mathbf{F} + \underbrace{\mathbf{d}(t)}_{\text{whatever the model got wrong}}. \quad (11)$$

This is the plant the QP optimises over: a double integrator driven by the task-space force, plus a disturbance. The crucial point for Section 2.4 is that $\mathbf{d}(t)$ is not an independent physical quantity — it is *defined* as whatever the model’s own cancellation fails to remove. Unmodelled friction goes straight into it.

The discrete plant and the QP. Discretising Eq. (11) over one tick with the error state $\mathbf{x} = [\mathbf{e}; \dot{\mathbf{e}}]$ gives

$$A_d = \begin{bmatrix} I & dt I \\ 0 & I \end{bmatrix}, \quad B_d(\mathbf{q}) = \begin{bmatrix} \frac{1}{2} dt^2 \boldsymbol{\Lambda}^{-1}(\mathbf{q}) \\ dt \boldsymbol{\Lambda}^{-1}(\mathbf{q}) \end{bmatrix}, \quad dt = 0.01 \text{ s}. \quad (12)$$

A_d is pose-independent and is built once; $B_d(\mathbf{q})$ is rebuilt every tick from the newly measured $\mathbf{q}$ and then held fixed across the horizon, which is the standard quasi-LPV treatment of a scheduled input matrix. Over a horizon of $N = 10$ steps the decision variable is the sequence of *centred* forces $\mathbf{v}_k = \mathbf{F}_k + \boldsymbol{\Lambda} \hat{\mathbf{d}}$ and the QP solved every tick by OSQP is

$$\min_V (\Phi \mathbf{x}_0 + \Gamma V)^\top \bar{Q} (\Phi \mathbf{x}_0 + \Gamma V) + V^\top \bar{R} V \quad \text{s.t.} \quad \begin{cases} -\mathbf{F}_{\max}^{\text{MPC}} \leq \mathbf{F}_k \leq \mathbf{F}_{\max}^{\text{MPC}}, & k = 0 \dots N-1, \\ -\tau_{\max} \leq \tau_{\text{base}} + \mathbf{J}_{xz}^\top \mathbf{F}_0 \leq \tau_{\max}, \end{cases} \quad (13)$$

with Φ built from A_d , Γ from $B_d(\mathbf{q})$, $\bar{Q} = \text{blkdiag}(Q, \dots, Q, \gamma Q)$, $Q = \text{diag}(q_{\text{pos}} I, q_{\text{vel}} I)$, $\bar{R} = \text{blkdiag}(r I)$, terminal weight $\gamma = 5$, force weight $r = 10$ and $\mathbf{F}_{\max}^{\text{MPC}} = 300 \text{ N}$. The actuator-feasibility constraint is enforced on the first predicted step only. Only the first element of the optimal sequence is applied and the loop is re-solved on the next tick.

The disturbance observer. $\mathbf{d}(t)$ in Eq. (11) is not measured, so it is estimated. The observer augments the error state with the disturbance as a random walk,

$$\mathbf{z} = \begin{bmatrix} \mathbf{e} \\ \dot{\mathbf{e}} \\ \mathbf{d} \end{bmatrix}, \quad \mathbf{z}_{k+1} = \underbrace{\begin{bmatrix} A_d & B \\ 0 & I \end{bmatrix}}_{A_a} \mathbf{z}_k + \underbrace{\begin{bmatrix} B \\ 0 \end{bmatrix}}_{B_a} \mathbf{u}_k + \mathbf{w}_k, \quad \mathbf{y}_k = \underbrace{\begin{bmatrix} I & 0 & 0 \end{bmatrix}}_C \mathbf{z}_k + \mathbf{v}_k, \quad (14)$$

where $B = [\frac{1}{2} dt^2 I; dt I]$ and the measurement $\mathbf{y}$ is the task-space position error itself. The process and measurement covariances are

$$Q_{\text{obs}} = \text{diag}(10^{-7} I, 10^{-5} I, q_{\hat{\mathbf{d}}} I), \quad R_{\text{obs}} = r_y I, \quad q_{\hat{\mathbf{d}}} = 10^{-4}, \quad r_y = 5 \cdot 10^{-5}, \quad (15)$$

and the recursion is the textbook predict/correct pair, with one addition: the disturbance block of $\mathbf{z}$ is clipped to $\pm \hat{d}_{\max}$ after every update, as anti-windup. Note that the position and velocity entries of Q_{obs} are fixed constants, so $q_{\hat{\mathbf{d}}}$ and r_y are *not* redundant with one another — $q_{\hat{\mathbf{d}}}$ sets how fast the disturbance state is allowed to move relative to the kinematic states, r_y sets how much the measurement is trusted. The estimate enters the QP as the force-equivalent $\boldsymbol{\Lambda} \hat{\mathbf{d}}$, which is why the centred variable $\mathbf{v}_k$ appears in Eq. (13).

Figure 5 shows how the pieces connect.

Two implementation facts. Both are needed later and both are taken from the logs, not from the design intent.

1. **Neither constraint in Eq. (13) was ever active in this study.** Torque saturation occurred on 0.00 % of ticks and the peak commanded force was $|\mathbf{F}| = 1.41 \text{ N}$ against a 300 N box. The QP is solved every tick, but in this operating regime it always returns its unconstrained optimum. This is an empirical statement about the present regime, not a claim that MPC reduces to PD in general — as soon as a constraint binds, the equivalence of Section 4.1 fails.

MPC torque-control loop, 100 Hz. The QP solves for the task-space force directly; $\Lambda^{-1}(q)$ sits inside the discrete input matrix $B_d(q)$, not after the solve.

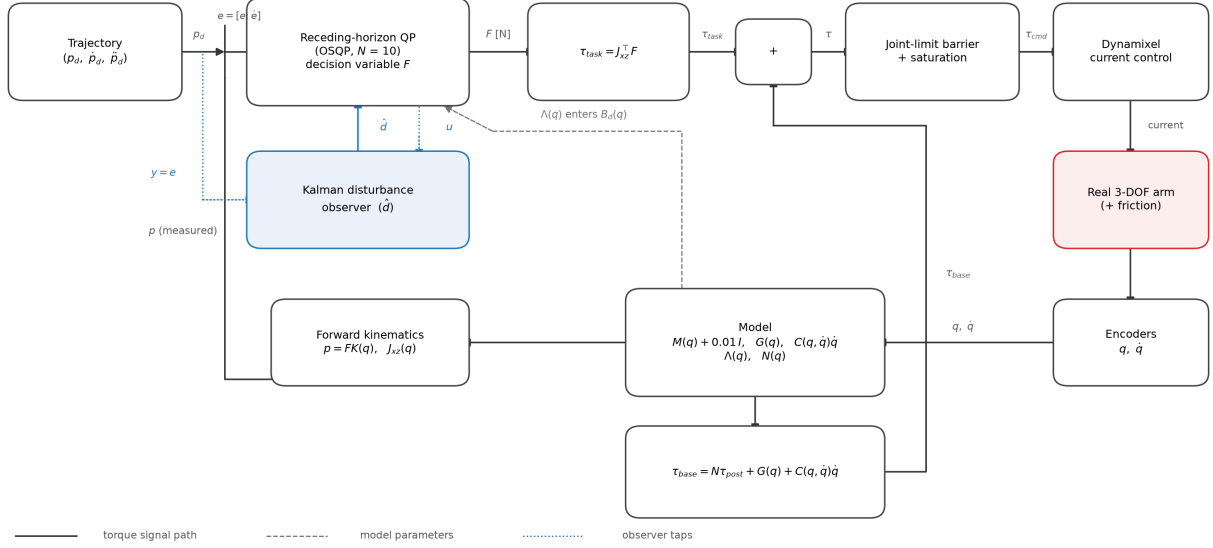


Figure 5: The MPC torque-control loop at 100 Hz.

2. **The cost contains no preview of the future reference.** It acts on the free response $\Phi \mathbf{x}_0$ of the error state; $\ddot{\mathbf{p}}_d$ appears only in the constant term of the torque constraint. The horizon shapes the feedback gain, it does not supply feed-forward.

3.2 Verifying the observer

The observer is the one component of the loop whose output cannot be checked against a measurement, since $\mathbf{d}(t)$ has no sensor. What can be checked is *convergence*: started from a deliberately wrong initial state, does $\mathbf{z}$ return to the same trajectory it would have followed anyway?

A hold run was made on hardware with both blocks of $\mathbf{z}$ seeded away from the truth: the position-error block at +50 mm on both task axes and the disturbance block at +0.30 m/s² on both. Everything else — the trajectory, the gains, the covariances of Eq. (15) — was left at its normal value. Figure 6 shows the result.

The two blocks converge by different routes and the difference is structural rather than incidental:

- The position-error block is what C measures directly (Eq. (14)), so the correction step removes the error in a **single 10 ms tick**: the first innovation is −49.7 mm and the estimate is back to 0.34 mm on the next sample. This verifies that the correction step is wired correctly, but it is not a demanding test of the filter’s own dynamics.
- The disturbance block is *not* measured. It can only be inferred through the process model, from the accumulated discrepancy between predicted and measured position error, at a rate set by $q_{\hat{d}}$. It therefore decays much more slowly, reaching $|\hat{d}_x| < 0.05$ m/s² after **1.22 s** — about 122 ticks — before settling onto the value the model residual actually requires.

Panel (c) records the cost of the wrong prior, which is not negligible: the controller acts on the seeded estimate, so the arm is driven away from its hold pose to a peak error of 20.99 mm at

Observer convergence from a deliberately wrong initial state (hardware, hold trajectory)

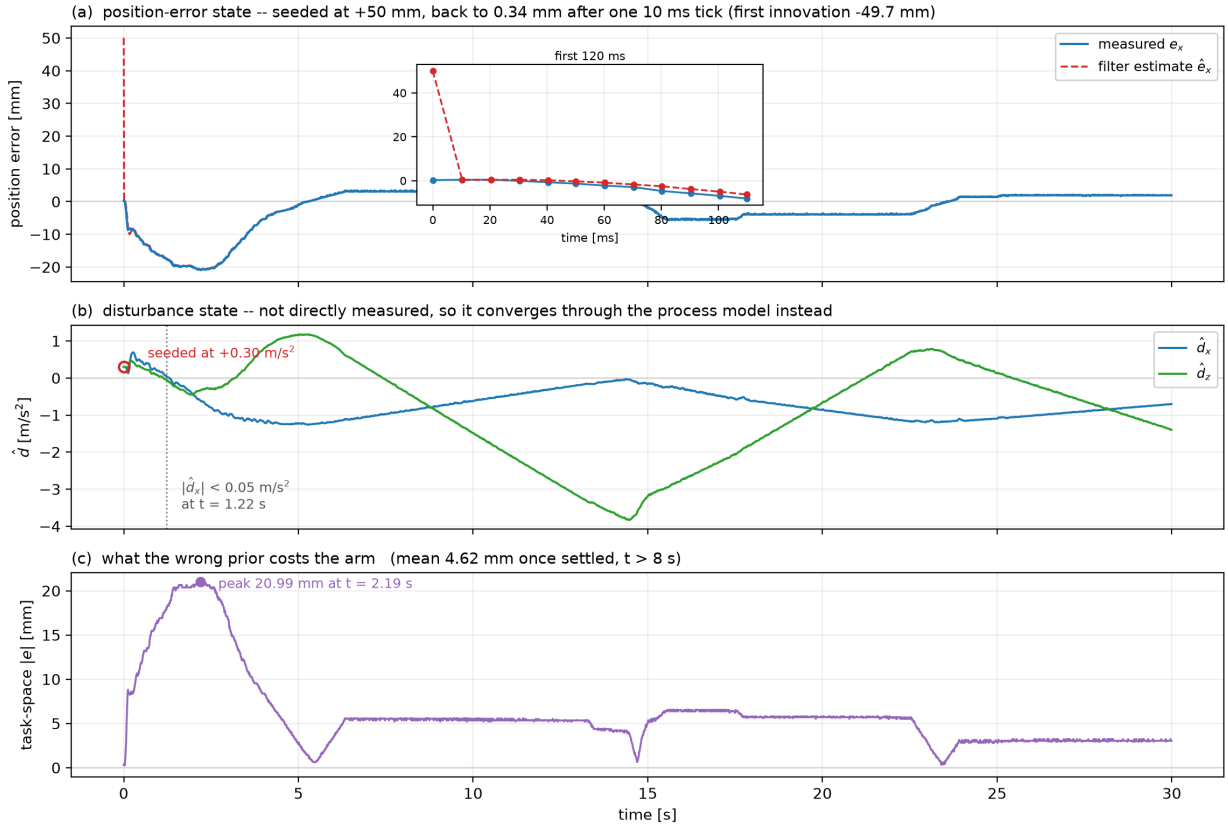


Figure 6: Observer convergence from a deliberately wrong initial state, hardware, hold trajectory. (a) the position-error block, seeded at +50 mm; (b) the disturbance block, seeded at $+0.30 \text{ m/s}^2$; (c) the resulting excursion of the arm itself.

$t = 2.19$ s before recovering. That the loop recovers at all from a 50 mm error in its own state estimate is the practical content of this test. The residual 4.62 mm mean after $t = 8$ s is the ordinary hold accuracy of this arm at the low stiffness used for the test, not a remnant of the injected bias.

The switches used here (`observer_pos_error_init` and `observer_d_hat_init`) are optional configuration keys and are disabled for every other result in this report.

3.3 Task-space PID

The baseline computes the task-space force directly from the same error signal the MPC uses, with no model term of any kind:

$$\mathbf{F} = -K_p^{\text{PID}} \mathbf{e} - K_i^{\text{PID}} \int_0^t \mathbf{e} d\sigma - K_d^{\text{PID}} \dot{\mathbf{e}}, \quad \left\| K_i^{\text{PID}} \int \mathbf{e} \right\|_{\infty} \leq I_{\max}^{\text{PID}}, \quad \|\mathbf{F}\|_{\infty} \leq F_{\max}^{\text{PID}}, \quad (16)$$

after which the same $\boldsymbol{\tau} = \mathbf{J}_{xz}^{\top} \mathbf{F} + \boldsymbol{\tau}_{base}$ mapping of Eq. (10) is applied. The `robot` and `friction_comp` blocks of its configuration are byte-identical to the MPC's, so the two controllers differ only in how $\mathbf{F}$ is produced. In particular both receive the same gravity and Coriolis feed-forward; the comparison in Section 6.5 is between the two feedback laws, not between a model-based and a model-free controller.

The verified settings are $K_p^{\text{PID}} = 300$, $K_i^{\text{PID}} = 100$, $K_d^{\text{PID}} = 6.37$, $I_{\max}^{\text{PID}} = 0.25$ and $F_{\max}^{\text{PID}} = 3.0$ N at the first posture and $K_p^{\text{PID}} = 80$, $K_i^{\text{PID}} = 20$, $K_d^{\text{PID}} = 6.37$ at the second. Two consequences of Eq. (16) matter for the comparison:

- **The integral authority has a hard floor.** The measured breakaway force of this arm is 0.94 N, so unless $K_i^{\text{PID}} \cdot I_{\max}^{\text{PID}} > 0.94$ N the integrator saturates before it can break stiction and the arm stalls short of the setpoint. Any run violating this is void and the settings above are chosen to clear it.
- **Lowering the gain cuts both paths at once.** K_p^{PID} and K_i^{PID} are the controller's only two routes to the setpoint and detuning for stability shrinks them together. The MPC has a second, separate channel in $\Lambda \hat{\mathbf{d}}$, which is what Section 6.5 finds to be the decisive structural difference between them.

3.4 Time-delayed Control

TDC is implemented in its simplest published form (Memmer, Eq. 3.8/4.1), in *joint space*, with no TDCGE gradient-estimator term, no friction feed-forward and no gravity feed-forward:

$$\boldsymbol{\tau}(t) = \bar{D} \left(\ddot{\mathbf{q}}_d(t) + K_d^{\text{TDC}} \dot{\mathbf{e}}_q(t) + K_p^{\text{TDC}} \mathbf{e}_q(t) \right) + \boldsymbol{\tau}(t - \lambda) - \bar{D} \ddot{\mathbf{q}}(t - \lambda), \quad \mathbf{e}_q = \mathbf{q}_d - \mathbf{q}, \quad (17)$$

where λ is one control period and $\bar{D}$ is a constant diagonal nominal inertia,

$$\bar{D} = \text{diag}(2.4264, 2.7750, 0.3674) \cdot 10^{-4} \quad [\text{kg m}^2].$$

Two choices that the thesis's hardware chapter leaves open are made explicitly here:

- $\dot{\mathbf{e}}_q$ is the backward difference of the joint error $\mathbf{e}_q$, not $\dot{\mathbf{q}}_d - \dot{\mathbf{q}}_{\text{meas}}$, following the thesis's own remark (4.2.1) that differentiating the error gives better results than subtracting a noisy measured velocity;

- $\ddot{\mathbf{q}}(t - \lambda)$ has no sensor on this hardware, so it is obtained by backward-differencing the measured joint velocity.

Because TDC acts in joint space, the task-space step reference is converted by inverse kinematics before being handed to the controller. The gains used are $K_p^{\text{TDC}} = 100$, $K_d^{\text{TDC}} = 160$ ($\zeta = 8.0$) at 100 Hz, identical in simulation and on hardware.

4 Method

4.1 Equivalent task-space gain mapping

This section is what allows the MPC and the PID to be compared at matched stiffness rather than merely side by side, so it is developed in full. Its scope is those two controllers only: TDC has no task-space (K_p, K_d) and is therefore outside it, for the reasons set out in Section 4.5.

The problem. The two task-space controllers are not tuned through comparable quantities. The PID is tuned directly in K_p^{PID} [N/m] and K_d^{PID} [Ns/m]: those numbers *are* the stiffness and damping it applies. The MPC is tuned through the LQR weights q_{pos} , q_{vel} and r , which are dimensionless and have no direct physical reading. Without a bridge between the two, statements such as “the MPC is more accurate than the PID” are uninterpretable, because nothing rules out that the two were simply run at very different stiffnesses.

The precondition. Everything below rests on one measured fact from Section 3.1: no constraint was ever active. Torque saturation 0.00 %; peak $|\mathbf{F}| = 1.41$ N against a 300 N box. Therefore OSQP returns, every tick, the minimiser of an *unconstrained* convex quadratic — and that minimiser is available in closed form.

The derivation. Stacking the horizon, the predicted error trajectory is $\mathbf{X} = \Phi \mathbf{x}_0 + \Gamma \mathbf{V}$, where Φ is built from A_d alone, Γ from $B_d(\mathbf{q})$ of Eq. (12) and $\mathbf{V}$ is the stacked sequence of centred forces $\mathbf{v}_k = \mathbf{F}_k + \Lambda \dot{\mathbf{d}}$. The cost is

$$J = (\Phi \mathbf{x}_0 + \Gamma \mathbf{V})^\top \bar{Q} (\Phi \mathbf{x}_0 + \Gamma \mathbf{V}) + \mathbf{V}^\top \bar{R} \mathbf{V}, \quad (18)$$

with $\bar{Q} = \text{blkdiag}(Q, \dots, Q, \gamma Q)$, $Q = \text{diag}(q_{pos}I, q_{vel}I)$ and $\bar{R} = \text{blkdiag}(rI)$. Setting $\partial J / \partial \mathbf{V} = 0$,

$$\underbrace{(\Gamma^\top \bar{Q} \Gamma + \bar{R})}_H \mathbf{V} = -\Gamma^\top \bar{Q} \Phi \mathbf{x}_0 \quad \implies \quad \mathbf{V}^* = -H^{-1} \Gamma^\top \bar{Q} \Phi \mathbf{x}_0, \quad (19)$$

which is *linear* in the current error state. Receding-horizon execution applies only the first block of $\mathbf{V}^*$, so with $\mathbf{K} := [H^{-1} \Gamma^\top \bar{Q} \Phi]_{1:n}$ the force actually commanded this tick is

$$\boxed{\mathbf{F}_0 = -\mathbf{K}(\Lambda(\mathbf{q})) [\mathbf{e}; \dot{\mathbf{e}}] - \Lambda(\mathbf{q}) \hat{\mathbf{d}}} \quad (20)$$

The gain matrix has the right units already. With the state ordered $[e_x, e_z, \dot{e}_x, \dot{e}_z]$, $\mathbf{K}$ is 2×4 . At the final first-posture configuration ($q_{pos} = 120000$, $q_{vel} = 0$, $r = 10$):

$$\mathbf{K} = \begin{bmatrix} 101.392 & 0.585 & 8.269 & -0.005 \\ 0.585 & 106.135 & -0.005 & 8.230 \end{bmatrix}. \quad (21)$$

$\mathbf{F}$ is in newtons and $\mathbf{e}$ in metres, so $K_{11} = 101.39$ is already in N/m and $K_{13} = 8.269$ is already in Ns/m — no conversion is involved. These are exactly the two quantities the PID is given directly. The off-diagonal entries (0.585 and -0.005) are two orders of magnitude smaller than the diagonal, which is why quoting only the diagonal is justified.

Verification against the logs. The mapping is checked, not assumed. Reconstructing Eq. (20) from run `a41_pos`’s logged $\mathbf{q}$, $\mathbf{e}$, $\dot{\mathbf{e}}$ and $\hat{\mathbf{d}}$ reproduces the logged commanded force to a residual of **0.010 N** during dwells (against $|\mathbf{F}| \approx 0.7$ N) and 0.114 N during commanded moves, the latter difference coming from rebuilding $\mathbf{\Lambda}$ and the error state from sampled log data plus OSQP’s own 10^{-4} tolerance.

What the mapping requires. Three conditions must be stated with it. (i) No constraint may be active; once one binds, V^* is no longer the closed-form minimiser and the equivalence fails. (ii) $\mathbf{\Lambda}(\mathbf{q})$ is frozen across the horizon — this is the controller’s own quasi-LPV assumption, not an extra approximation introduced for the mapping. (iii) $\mathbf{K}$ depends on the arm posture through $\mathbf{\Lambda}(\mathbf{q})$, so “the gain” is a pose-dependent quantity; this is why the two postures of Section 4.4 occupy different points on the gain plane.

Property 1: K_p^{MPC} **saturates in q_{pos} .** The mapping is strongly non-linear and reading the weights as if they were gains is misleading. Computing the local slope $d \ln K_p^{\text{MPC}} / d \ln q_{\text{pos}}$ at the first posture with $q_{\text{vel}} = 0$:

q_{pos}	10^3	10^4	$3 \cdot 10^4$	10^5	$3 \cdot 10^5$	10^6
K_p^{MPC} [N/m]	1.32	12.67	36.36	89.75	173.59	276.57
local slope	0.99	0.95	0.86	0.68	0.47	0.38

At low weights the force penalty r dominates H and the gain is *linear* in q_{pos}/r (slope 0.99). Over the range actually used in this report, 10^5 to $3 \cdot 10^5$, the slope has already fallen to about 0.5: **tripling the weight buys only 1.9× the stiffness.**

Property 2: q_{vel} **does not simply add damping.** The LQR trades the two weights against each other, so raising q_{vel} raises K_d^{MPC} and lowers K_p^{MPC} . At fixed $q_{\text{pos}} = 120000$:

q_{vel}	0	1200	12000	120000
K_p^{MPC} [N/m]	101.39	69.78	30.59	6.78
K_d^{MPC} [Ns/m]	8.27	12.98	27.31	51.67

This has a concrete consequence for the tuning history of this project. Every gain sweep before the final ladder used the rule $q_{\text{vel}} = q_{\text{pos}}/10$, which ties K_d^{MPC} to K_p^{MPC} : it is then structurally impossible to lower the damping while keeping the stiffness. Setting $q_{\text{vel}} = 0$ is what decouples the two and only then could the stability boundary of Section 6.1 be located.

Both properties are shown in Figure 7, with the configurations actually used in this report marked on the curves.

Table 4 already shows the fact that motivates Section 6.5: at their verified operating points the PID commands roughly **three times** the stiffness of the MPC (300 vs 101 N/m). Any accuracy comparison that ignores this is not a comparison of the control laws.

Mapping the MPC's LQR weights onto the task-space gains the PID uses directly ($r = 10$, $\gamma = 5$, horizon 10)

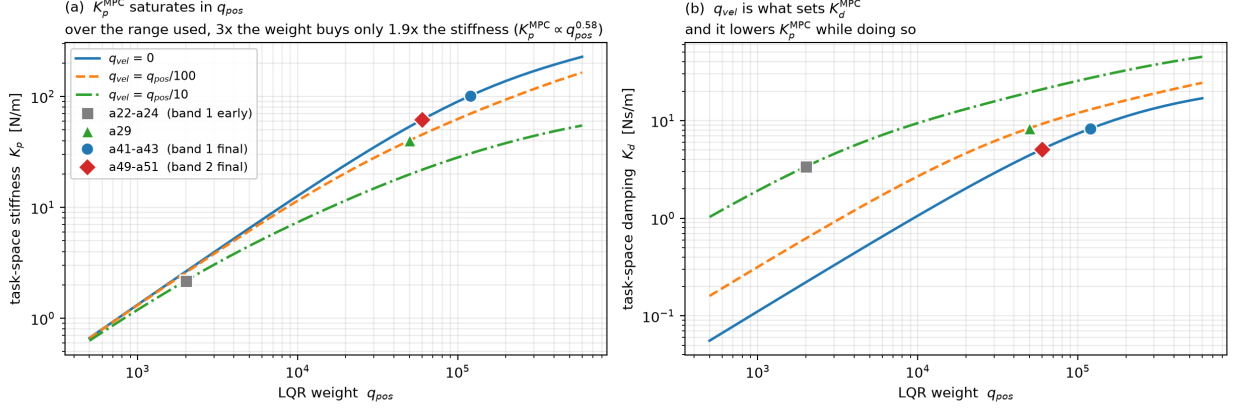


Figure 7: The equivalent task-space gain mapping. (a) K_p^{MPC} against q_{pos} for three q_{vel} rules; (b) K_d^{MPC} for the same. The markers are the configurations used in this report.

Table 4: Every configuration in this report, expressed in the same task-space units. MPC entries are obtained from Eq. (20); PID entries are its tuning parameters as given.

Controller	Configuration	q_{pos}	q_{vel}	K_p [N/m]	K_d [Ns/m]
MPC	a22--a24, posture 1 (early)	2000	200	2.18	3.38
MPC	a29, posture 1	50000	500	40.05	8.38
MPC	a41--a43, posture 1 (final)	120000	0	101.39	8.27
MPC	a49--a51, posture 2 (final)	60000	0	61.49	5.08
PID	C18--C20, posture 1 (final)	—	—	300.00	6.37
PID	E4,E6,E7, posture 2 (final)	—	—	80.00	6.37

4.2 The loop-gain delay criterion

The observed divergence mode on this arm is an oscillation at $f_{\text{crit}} \approx 8$ Hz. The round-trip loop delay is $T_d \approx 30$ ms, made up of ≈ 5 ms zero-order hold (half the 10 ms period), ≈ 8 ms QP solve, ≈ 10 ms actuation, plus the XM430’s internal velocity filter. At $\omega = 2\pi f_{\text{crit}} = 50.27$ rad/s, the magnitude of the loop transfer through the task-space double integrator is

$$|L| = \frac{\sqrt{K_p^2 + (\omega K_d)^2}}{\Lambda_{xx} \omega^2}. \quad (22)$$

Because $\omega \approx 50$, the term ωK_d dominates K_p unless K_d is very small: **stability on this arm is governed by K_d , not by K_p** . Evaluated at the four verified operating points, Eq. (22) gives (here and in Figure 12, Λ and hence K_p , K_d are evaluated at one nominal pose per band rather than along the measured trajectory, so all $|L|$ values carry an uncertainty of a few per cent from the within-band variation reported in Section 6.5; this is far below the spacing of the conclusions drawn from them)

Configuration	K_p [N/m]	K_d [Ns/m]	$ L $
MPC, posture 1	101.39	8.27	0.255
PID, posture 1	300.00	6.37	0.261
MPC, posture 2	61.49	5.08	0.156
PID, posture 2	80.00	6.37	0.195

Two structurally different controllers, tuned independently and differing threefold in stiffness, land within 2% of each other in $|L|$ at the first posture. This is the single most compact statement of the report’s first result.

4.3 Repeatability protocol

Every headline number in this report is **three separate runs of eight commanded steps each**, $n = 24$. This protocol was adopted after a single-run gain sweep badly overstated the PID’s ceiling: settings that completed once failed on repetition (at K_p^{PID} 400 and 420 only 2 of 3 and 1 of 3 runs completed; at 350, 1 of 2; at 320, 0 of 1). Only $K_p^{\text{PID}} = 300$ with $K_i^{\text{PID}} = 100$ held 3 of 3 and that is the configuration reported. Results obtained from a single run are labelled $n = 8$ wherever they appear and are never used as conclusions.

4.4 Trajectory, postures and metrics

The reference is a sequence of **eight ± 20 mm steps** along the task-space x axis, the first at $t = 5$ s and 6.5 s apart, for a total run length of 57 s. Two **postures** are used, both starting from the same pose ($x = -30.4$ mm) and each exploring one 20 mm interval on one side of it:

- **posture band 1:** $x \in [-30.4, -10.4]$ mm;
- **posture band 2:** $x \in [-50.4, -30.4]$ mm.

Each band contains equal numbers of outbound and return moves, so the difference between them is **not** direction of travel; it is arm configuration and with it the gravity loading and friction the controller must overcome. Figure 8 shows both bands in task and joint space.

The metrics are defined once and used identically in simulation and on hardware:

The step reference used throughout, in both pose bands

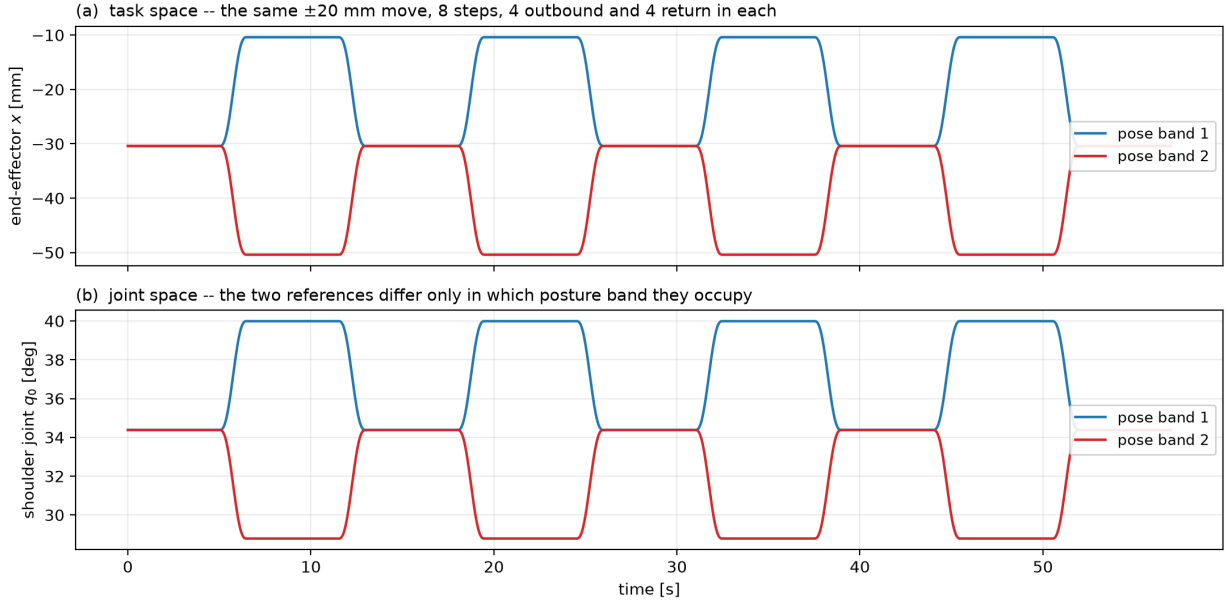


Figure 8: The step reference, shown for both posture bands in task space and in joint space.

- **rise time:** 10%–90% of the commanded travel;
- **overshoot:** peak excursion beyond the commanded travel, in per cent of that travel;
- **settling time:** last exit from a ± 3.0 mm band about the new setpoint;
- **steady-state error** and $|e_x|$: mean signed and mean absolute error over the *last 2.0 s* of each dwell, measured **along the stepped (x) axis**.

The single-axis definition of $|e_x|$ was adopted on 2026-08-28 to match the tooling of the sibling TDC project; the previously used two-dimensional error is systematically larger and is not quoted here.

4.5 What the comparison does not equalise

Sections 4.1–4.4 state what is held equal: one reference trajectory, one loop rate, one set of metric definitions, one repeatability protocol and two postures for every controller. Three things are *not* equalised and they bound what the comparison can establish.

TDC does not act in the same space. The MPC and the PID both compute a task-space force and map it to joint torque through $J_{xz}^\top$. TDC computes joint torque directly from a joint-angle error, so its input is three joint angles and the task-space reference has to be converted by inverse kinematics before it reaches the controller. Two consequences follow:

- **There is no gain parameterisation common to all three.** The mapping of Section 4.1 exists because the MPC’s first-step solution and the PID’s control law are both of the form $\mathbf{F} = -K_p \mathbf{e} - K_d \dot{\mathbf{e}}$ in the same units. TDC has no task-space (K_p, K_d) at all, so it cannot

be placed on the gain plane of Section 6.1 and the loop-gain criterion of Section 4.2 does not apply to it. The MPC and the PID can therefore be compared *at matched stiffness*; TDC can only be compared on outcome metrics.

- **The redundant degree of freedom is used differently.** Three joints serve a two-dimensional task (Section 2.1). The two task-space controllers command a force that the dynamically consistent inverse distributes over all three joints, so a joint that is temporarily unable to deliver its share can be partly compensated by the other two. TDC tracks each joint independently against a diagonal $\bar{D}$ and has no such redistribution.

The three did not receive the same tuning effort, or the same kind of tuning. The MPC and the PID were driven to their *stability* ceilings: 92 hardware runs mapped the boundary of Section 6.1, and each controller’s reported configuration is the highest gain that completed three runs out of three in that posture band. TDC’s $\bar{D}$ was found by a 12-run *performance* sweep (Section 6.2) which never encountered instability at all. This asymmetry is itself one of the findings — on this hardware the two task-space controllers are gain-limited and TDC is not — but it does mean the comparison is between three controllers each tuned as far as its own binding constraint allowed, not between three controllers given equal numbers of attempts.

The simulation could not be made to agree. MPC and PID use byte-identical configurations in Chapters 5 and 6. TDC cannot: its hardware configuration diverges in simulation (Section 6.3), so the simulated TDC of Section 5.2 is the earlier $\bar{D}$. Where TDC’s simulated and measured numbers are placed side by side, they are therefore not the same configuration and are labelled as such.

What survives all three. The outcome metrics of Section 4.4 are measured on the same trajectory, at the same rate, with the same definitions, from three repeats each. Statements of the form “controller A held the setpoint to x mm where controller B held it to y mm, at this arm and this task” are therefore supported. Statements of the form “method A is better than method B” are not and the report does not make them.

5 Simulation Results

5.1 Simulation design

All simulation runs use the trajectory of Section 4.4 at the first posture: 100 Hz, 57 s, 5700 samples, eight ± 20 mm steps. Each controller uses the same configuration it later uses on hardware. The TDC reference is converted to joint space by inverse kinematics before being handed to the controller, as it is on hardware.

That the three controllers share *one* reference is verified rather than assumed: the logged p_d traces agree to $8.6 \cdot 10^{-11}$ m across the three controllers and to 10^{-7} mm across the three loop rates of Section 5.3 after resampling. The figure-generating script re-checks this at load time and refuses to plot if it ever fails.

Two limitations must be kept in view when reading this section. The simulation contains **no Coulomb friction** and **no loop delay**. Parameters that exist specifically to break through stiction — the PID’s K_i^{PID} , the observer’s $q_{\hat{d}}$ — therefore cannot be calibrated here and the stability boundary of Section 6.1 does not appear at all. In addition, the simulation is deterministic, so it carries no information about repeatability.

5.2 The three controllers

Simulation, 100 Hz -- MPC, task-space PID and TDC on the identical step reference

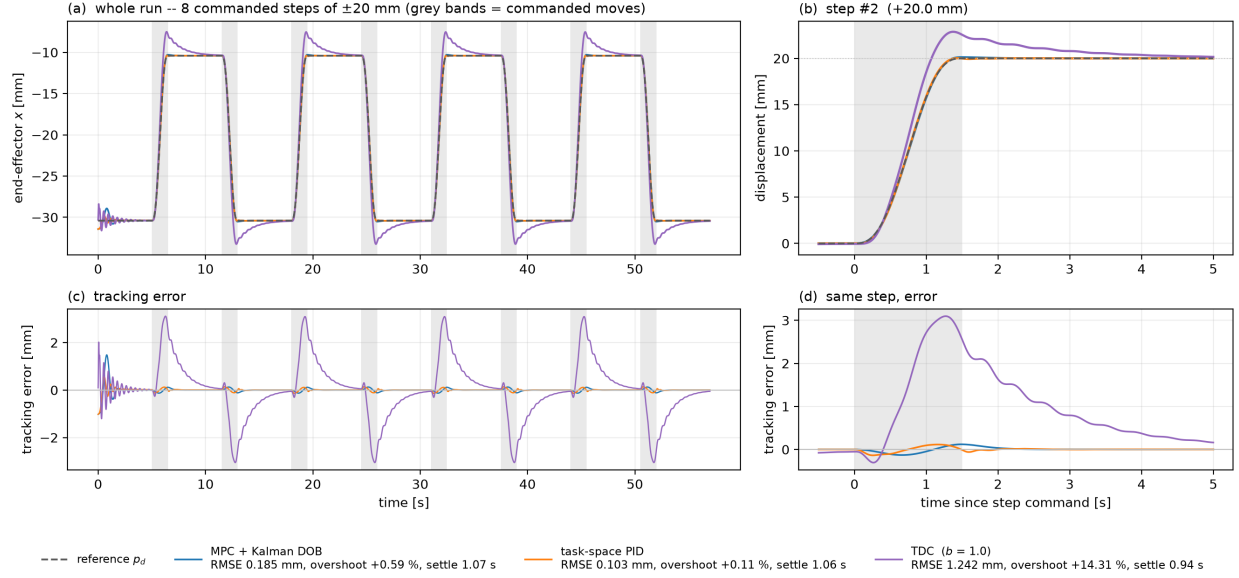


Figure 9: Simulation at 100 Hz: MPC, task-space PID and TDC on the identical step reference. Left: whole run. Right: one step in detail.

Table 5: Simulation, 100 Hz, eight steps.

Controller	RMSE [mm]	rise [s]	overshoot [%]	settling [s]
MPC + Kalman DOB	0.185	0.750	+0.59	1.070
Task-space PID	0.103	0.750	+0.11	1.060
TDC ($b = 1.0$)	1.242	0.600	+14.31	0.940

The two task-space controllers are essentially indistinguishable in simulation (Table 5): both track the reference to within a few tenths of a millimetre, with negligible overshoot. TDC is the fastest to rise (0.600 s) but overshoots by 14.3% and carries an order of magnitude more RMSE.

The overshoot is not a defect of the TDC law as such; it is a consequence of running it at 100 Hz, which Section 5.3 makes explicit. Because the simulation lacks friction and delay, these numbers should be read as a check that all three laws are correctly implemented and stable, not as a prediction of hardware ranking — Section 6 shows the ranking changes.

5.3 TDC at three loop rates

The gains are *not* shared across the three rates and cannot be: a tuning that is stable at 500 Hz diverges outright at 250 and 100 Hz. Each rate is therefore represented by its own lowest-RMSE tuning. The gains quoted in Table 6 were recovered from the logs themselves by least squares on Eq. (17), rather than copied from configuration files.

Overshoot falls monotonically with rate, $14.31 \rightarrow 3.61 \rightarrow 1.04\%$ and RMSE with it. Rise time moves the other way ($0.600 \rightarrow 0.732$ s), which is a direct consequence of each rate needing its own

TDC in simulation at three loop rates -- each at its own lowest-RMSE tuning (gains do not transfer across dt)

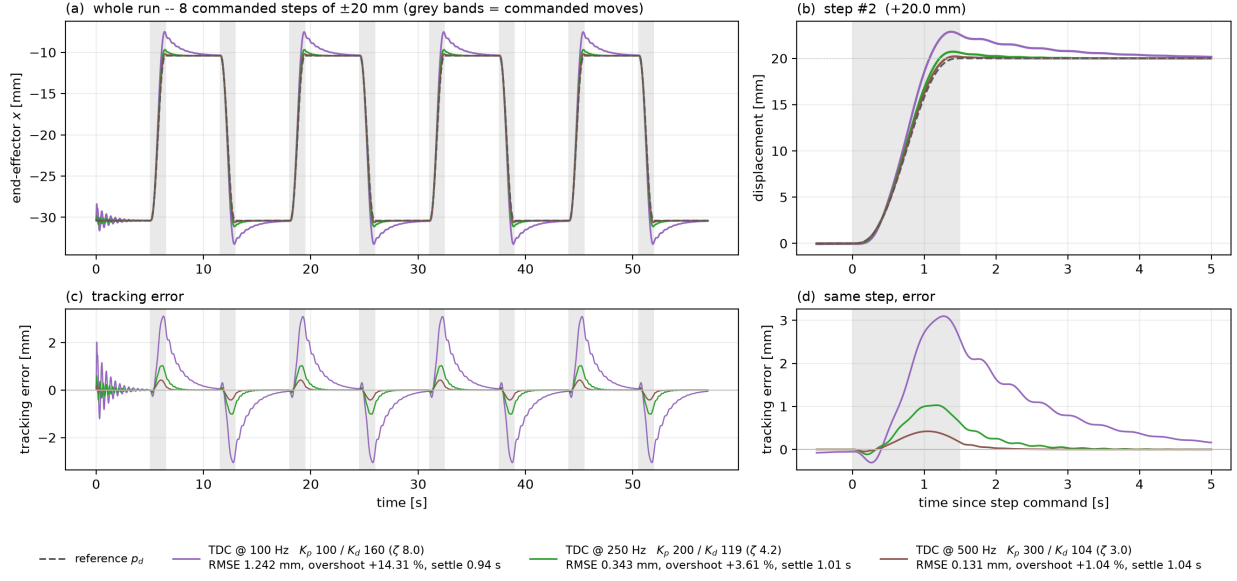


Figure 10: TDC in simulation at 100, 250 and 500 Hz on the identical reference, each at its own lowest-RMSE tuning.

Table 6: TDC in simulation at three loop rates, each at its own lowest-RMSE tuning.

Rate	K_p^{TDC}	K_d^{TDC}	ζ	RMSE [mm]	overshoot [%]	settling [s]
100 Hz	100	160.0	8.0	1.242	+14.31	0.940
250 Hz	200	119.4	4.2	0.343	+3.61	1.012
500 Hz	300	103.9	3.0	0.131	+1.04	1.040

tuning rather than a property of the rate itself.

5.4 A countermeasure for the 100 Hz overshoot

The 100 Hz overshoot can be attacked without changing the loop rate. In Eq. (17), $\dot{e}_q$ is the backward difference of the joint error, which is identical to $\dot{q}_d - \dot{q}_{\text{meas}}$. Applying a *setpoint weight* b to the reference half only,

$$\dot{q}_{\text{meas}} = \frac{q(t) - q(t-1)}{dt}, \quad \dot{q}_d = \dot{e}_q + \dot{q}_{\text{meas}}, \quad \dot{e}_q \leftarrow b \dot{q}_d - \dot{q}_{\text{meas}}, \quad (23)$$

recovers the original law exactly at $b = 1$ and at $b = 0$ leaves pure damping on the measured velocity. Both halves still come from the same position-difference signal, so this introduces *no new noise source*; it is not the “derivative on measured velocity” the thesis warns against in 4.2.1.

The motivation is quantitative. At 100 Hz the gains required for stability are $\zeta = 8$, $K_d^{\text{TDC}} = 160$. A term decomposition over the commanded moves shows that $K_d^{\text{TDC}} \dot{e}_q$ then supplies about 70 % of the control effort (peak -4.49 rad/s^2), against 35 % at 500 Hz. With K_d^{TDC} that large, the $\dot{q}_d$ half acts as an implicit velocity feed-forward of gain 160 and drives the arm *past* the reference: the 100 Hz tracking error is positive and still growing while the commanded move is in progress, i.e. the arm leads rather than lags. Reducing b targets exactly that half.

TDC at 100 Hz -- derivative setpoint weighting b against the unmodified law ($b = 1$), simulation only

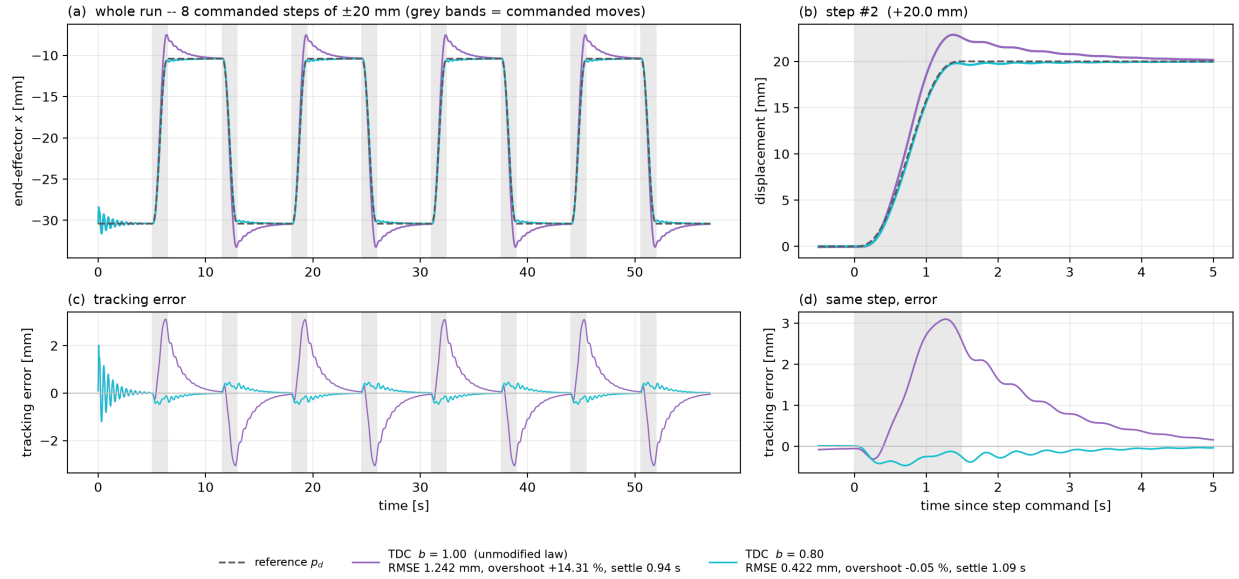


Figure 11: TDC at 100 Hz in simulation: derivative setpoint weighting $b = 0.80$ against the unmodified law $b = 1$.

At $b = 0.80$ the overshoot is removed entirely and RMSE falls by a factor of three, at the cost of 26 % slower rise and 15 % slower settling.

What happened to this route on hardware. With $b \neq 1$ the control law is no longer the one published in the thesis, so this was treated as a fallback rather than the primary route. It was nonetheless measured: one run of eight steps at $b = 0.80$ on hardware gives overshoot $-0.47 \pm 4.42 \%$,

Table 7: Effect of the derivative setpoint weight, simulation at 100 Hz.

	RMSE [mm]	rise [s]	overshoot [%]	settling [s]
$b = 1.00$ (unmodified law)	1.242	0.600	+14.31	0.940
$b = 0.80$	0.422	0.755	−0.05	1.085
$b = 0.75$	0.615	0.815	−0.14	1.140

settled error 1.339 mm and settling 3.045 s, against $+14.05 \pm 8.02\%$, 1.611 mm and 3.465 s for the unmodified law at the same $\bar{D}$. The overshoot is indeed removed, but the settled error is barely improved and the rise time doubles.

The route that was taken instead leaves the control law untouched and retunes $\bar{D}$, which the thesis leaves free (Section 6.2). That route reaches $+3.31 \pm 1.75\%$ overshoot, 0.395 mm settled error and 1.185 s settling, at $n = 24$ — better than $b = 0.80$ on every metric, without departing from Eq. (17). The simulation gives the opposite verdict on the two routes, which is the subject of Section 6.3.

6 Hardware Experimental Results

The hardware campaign uses **exactly the experimental design of Chapter 5**: the same trajectory, the same 100 Hz loop rate, the same 57 s run length, the same eight ± 20 mm steps. That the three controllers really are on one reference was checked for every reported run over the stepped portion ($t \geq 5$ s), resampling each log onto its band’s reference time grid so that the small differences in sample timing between runs (10.00 to 10.02 ms) do not enter the comparison. Across all eighteen runs the commanded x never departs from its band’s reference by more than **0.0032 mm** and the z reference is identical to 0.0000 mm.

The three TDC runs in each band are in fact the closest of all at 0.0002 mm — tighter than the MPC’s own repeats, which reach 0.0015 mm. The inverse-kinematics round trip that TDC’s reference has to pass through therefore costs nothing measurable.

Why a second posture band was added. The simulation of Chapter 5 has neither Coulomb friction nor loop delay, so a single arm posture cannot distinguish “this is a property of the controller” from “this is a property of this particular pose”. The second band (Section 4.4) is the same 20 mm move executed on the other side of the same start pose, with the same number of outbound and return steps but different gravity loading and different friction. It turned out to be the discriminating condition of the entire report.

The two task-space controllers had to be detuned for band 2. This is not a free choice and it is the reason their band-2 entries are not directly comparable with their band-1 entries. Carried over unchanged, the band-1 configurations *oscillate* in band 2: the boundary of Section 6.1 is systematically lower there and the highest configuration that completed three runs out of three drops from $q_{pos} = 120000$ to 60000 for the MPC (K_p^{MPC} 101.4 to 61.5 N/m, -39%) and from $K_p^{\text{PID}} = 300$ to 80 for the PID (-73%). Their band-2 numbers therefore contain two effects at once: the posture is harder *and* the controller was given less gain to meet it with. TDC needed no such reduction — its band-2 runs use the band-1 configuration unchanged — so its two bands isolate the posture effect alone.

6.1 The stability boundary: where the MPC and the PID can sit

Stability boundary of both controllers in one task-space gain plane

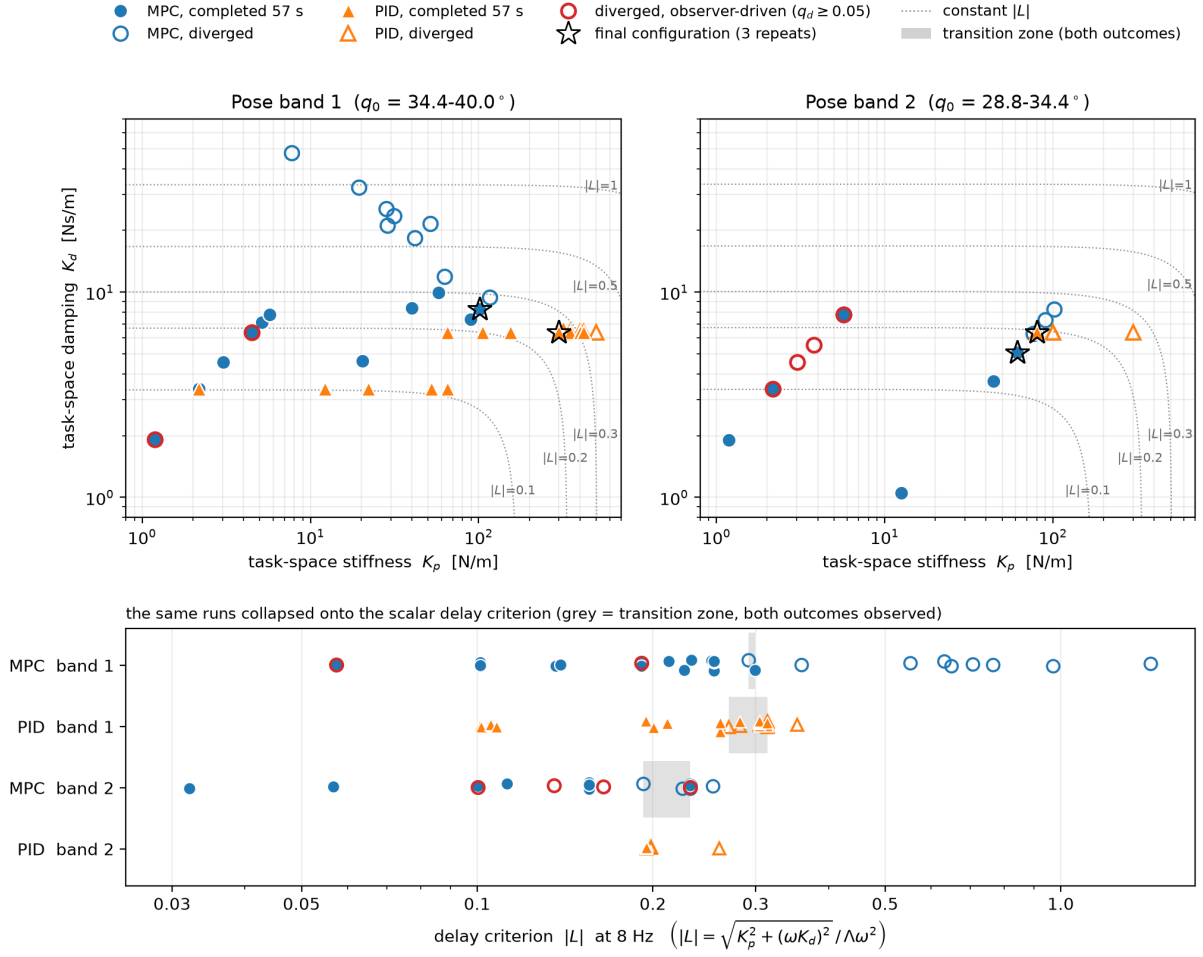


Figure 12: All 92 hardware runs of the two task-space controllers, both posture bands, on one equivalent (K_p, K_d) plane, with contours of $|L|$ from Eq. (22). TDC does not appear here: it has no task-space (K_p, K_d) (Section 4.5).

Ninety-two hardware runs were placed on the common gain plane of Section 4.1 — 56 completed and 36 diverged. Four observations:

1. The boundary runs approximately along a contour of $|L|$ and that contour is nearly *horizontal*: it is set by K_d , with K_p contributing very little. This is Eq. (22) read off the data.
2. **Both controllers obey the same boundary**, despite differing threefold in stiffness and having entirely different structures.
3. The boundary for posture band 2 is systematically lower, which is what forces the detuning described above.
4. A second, independent divergence mechanism is visible and is marked separately: 11 MPC divergences occurred *below* the boundary and every one of them had $q_{\hat{d}} \geq 0.05$ in the observer. With $q_{\hat{d}} \leq 0.015$ no run ever diverged below the boundary.

The boundary must be stated carefully, because all four lanes show an *overlap region* rather than a clean threshold:

	highest completed $ L $	lowest gain-driven divergence $ L $
MPC, band 1	0.299	0.291
PID, band 1	0.315	0.270
MPC, band 2	0.231	0.193
PID, band 2	0.195	0.195

The correct wording is therefore: $|L|$ *locates the boundary; inside the overlap region a third parameter decides*. It is not true that the boundary is determined by $|L|$ alone. A further caveat: the PID was only ever run at two K_d^{PID} values (3.37 and 6.37), so the vertical sampling of its lanes is sparse.

6.2 Choosing $\bar{D}$: where TDC can sit

TDC’s operating point is fixed by a different constraint, which is the reason it needs its own section rather than a point on Figure 12. In Eq. (17) the nominal inertia $\bar{D}$ multiplies the entire proportional–derivative bracket, so it scales the loop gain of each joint directly while K_p^{TDC} and K_d^{TDC} stay fixed. The thesis leaves $\bar{D}$ to be “determined by trial and error”, so choosing it is part of applying the method, not a modification of it.

Choosing $\bar{D}_0$: band 1, one run of eight steps per point ($K_p^{\text{TDC}} 100$, $K_d^{\text{TDC}} 160$, $b = 1.0$ throughout)

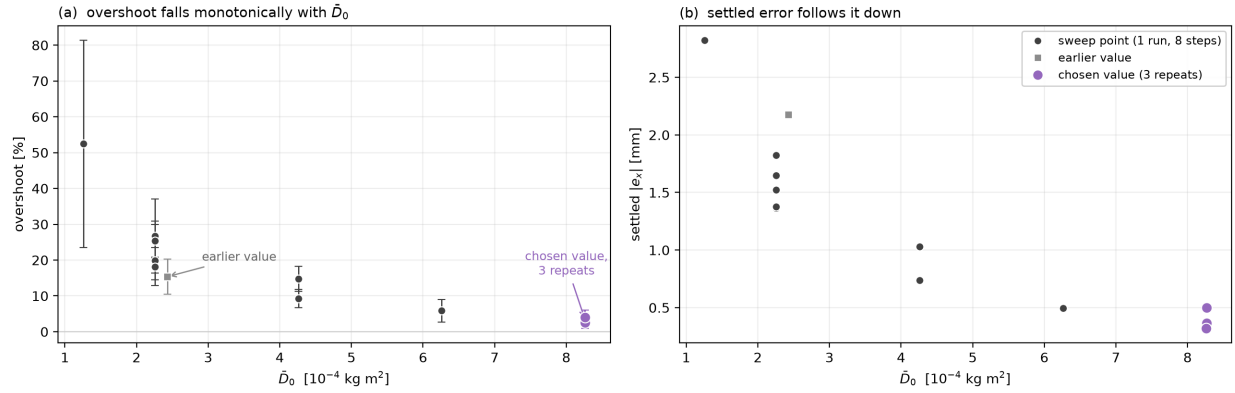


Figure 13: How $\bar{D}_0$ was chosen. Every point is one run of eight steps ($n = 8$), so this figure documents a selection procedure and is not itself a result; the report’s TDC conclusions rest only on the three repeats at the chosen value ($n = 24$).

Twelve runs on band 1, with $K_p^{\text{TDC}} = 100$, $K_d^{\text{TDC}} = 160$ and $b = 1.0$ held fixed throughout, swept $\bar{D}_0$ over a factor of 6.6. Overshoot falls monotonically across the whole range, from 52.4 % at $\bar{D}_0 = 1.26 \cdot 10^{-4}$ to 3.3 % at $8.26 \cdot 10^{-4}$ and the settled error follows it down from 2.82 to 0.37 mm. The chosen configuration is

$$\bar{D} = (8.26, 2.80, 1.52) \cdot 10^{-4} \text{ kg m}^2 \quad \text{against} \quad (2.43, 2.78, 0.37) \cdot 10^{-4}$$

for the earlier one — joint 1 raised 3.40 $\times$, joint 3 raised 4.14 $\times$, joint 2 essentially unchanged and 2.26 $\times$ in total. Since the CSV logs do not record control parameters, every $\bar{D}$ quoted here was

recovered from the logs themselves by least squares on Eq. (17), which reproduces the configured values to $R^2 = 1.000000$ on the clean runs.

Two properties of this selection matter for how much weight it can carry:

- **Each sweep point is a single run.** By the protocol of Section 4.3 that is not enough to conclude from. The sweep is reported as the procedure that produced a candidate; the candidate was then repeated three times and only those repeats are used as evidence.
- **The sweep was run on band 1 only.** The chosen value was transferred to band 2 without further adjustment and is used there unchanged. That it performs equally well at a posture it was never tuned on (Section 6.5) is the strongest available evidence that the value is not overfitted to one operating point.

Nothing in this sweep encountered instability. Where the MPC and the PID were stopped by the delay wall of Section 6.1, TDC was stopped by diminishing returns.

6.3 Overview of the verified configurations

Figures 14–16 show one representative run per controller per band, in the same layout as the simulation overviews so the two chapters can be read side by side.

Hardware overview -- MPC + Kalman DOB (dashed = reference, solid = measured; red band = startup ramp, grey bands = commanded moves)

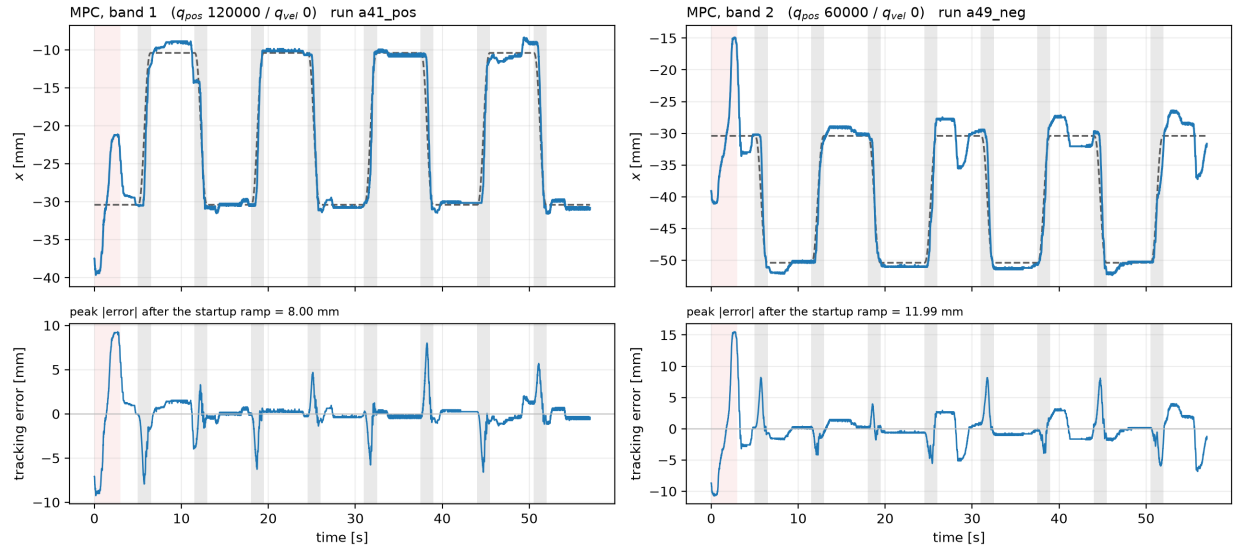


Figure 14: Hardware overview, MPC with Kalman disturbance observer, both posture bands.

Peak absolute error after the startup ramp, for the runs shown: MPC 8.00 and 11.99 mm at the two bands, PID 3.60 and 7.60 mm, TDC 4.05 and 3.38 mm. In every case the peak occurs in the step transient and all three settle close to the reference during the dwells. Peak transient error is therefore not a useful discriminator here and the report does not use it as one — the settled error of Section 6.5 is what separates the controllers.

The same TDC configuration in simulation. Running the configuration of Figure 16 — the same $\bar{D}$, K_p^{TDC} , K_d^{TDC} and reference — through the simulator of Chapter 5 gives Figure 17: it

Hardware overview -- task-space PID (dashed = reference, solid = measured; red band = startup ramp, grey bands = commanded moves)

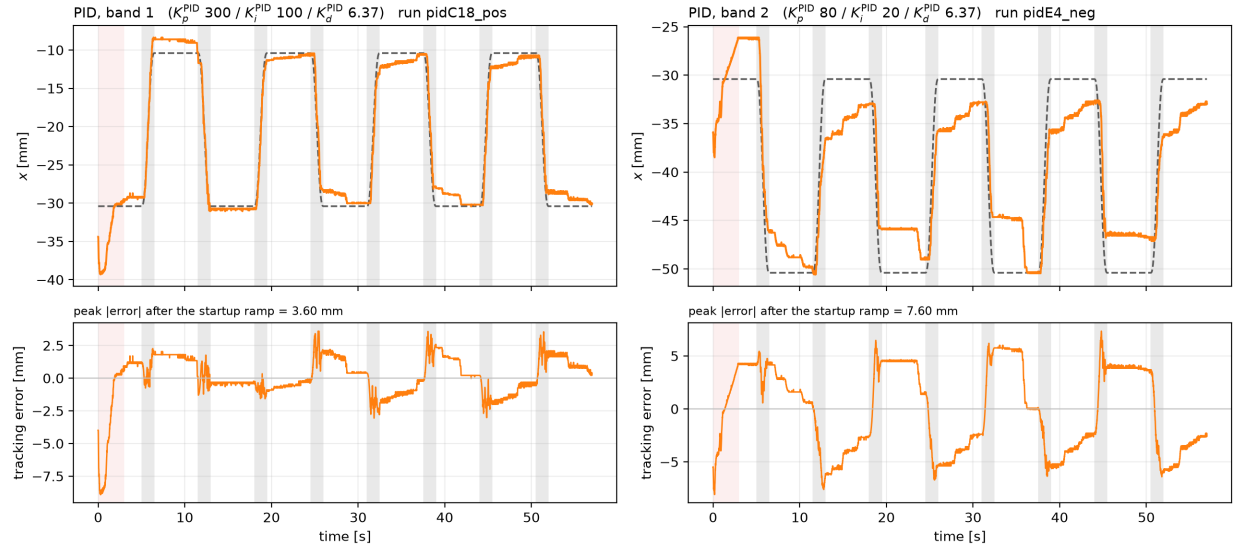


Figure 15: Hardware overview, task-space PID, both posture bands.

Hardware overview -- Time-delayed Control @ 100 Hz (dashed = reference, solid = measured; red band = startup ramp, grey bands = commanded moves)

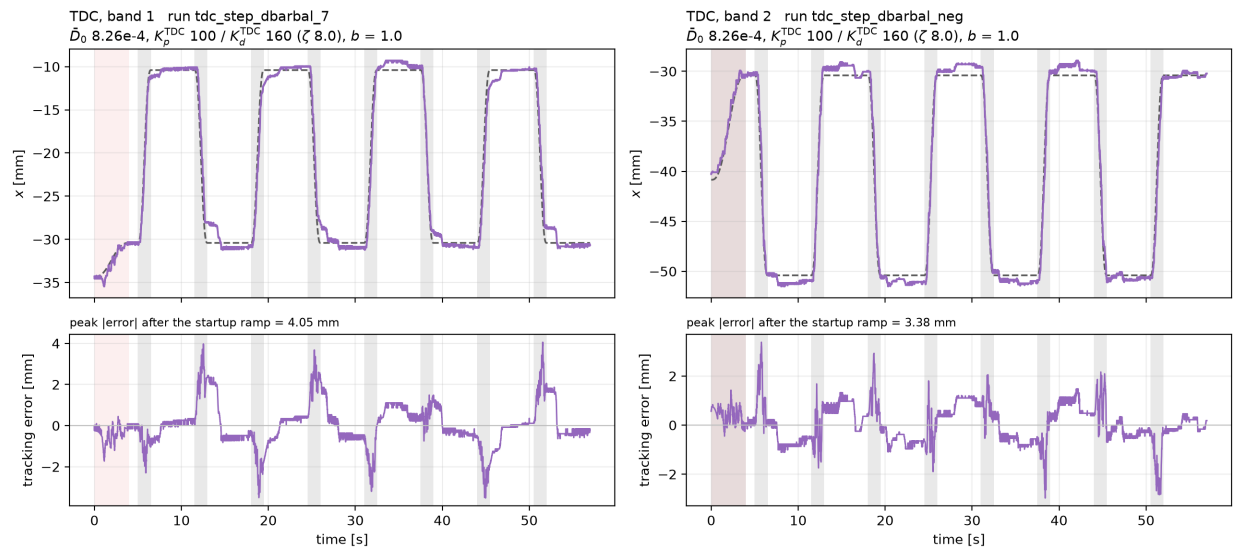


Figure 16: Hardware overview, Time-delayed Control at 100 Hz with the $\bar{D}$ of Section 6.2, both posture bands. The startup ramp is 4 s here against 3 s for the other two controllers.

leaves the hold pose within half a second, hits the joint-1 torque limit at $t = 0.45$ s and spends 46.9% of the run saturated, long before the first step is commanded at $t = 5$ s. The configuration was verified to be the intended one by recovering its parameters from the opening window, before any limit binds ($R^2 = 1.000000$).

The hardware-verified TDC configuration, run in simulation: it leaves the hold pose before the first step is commanded

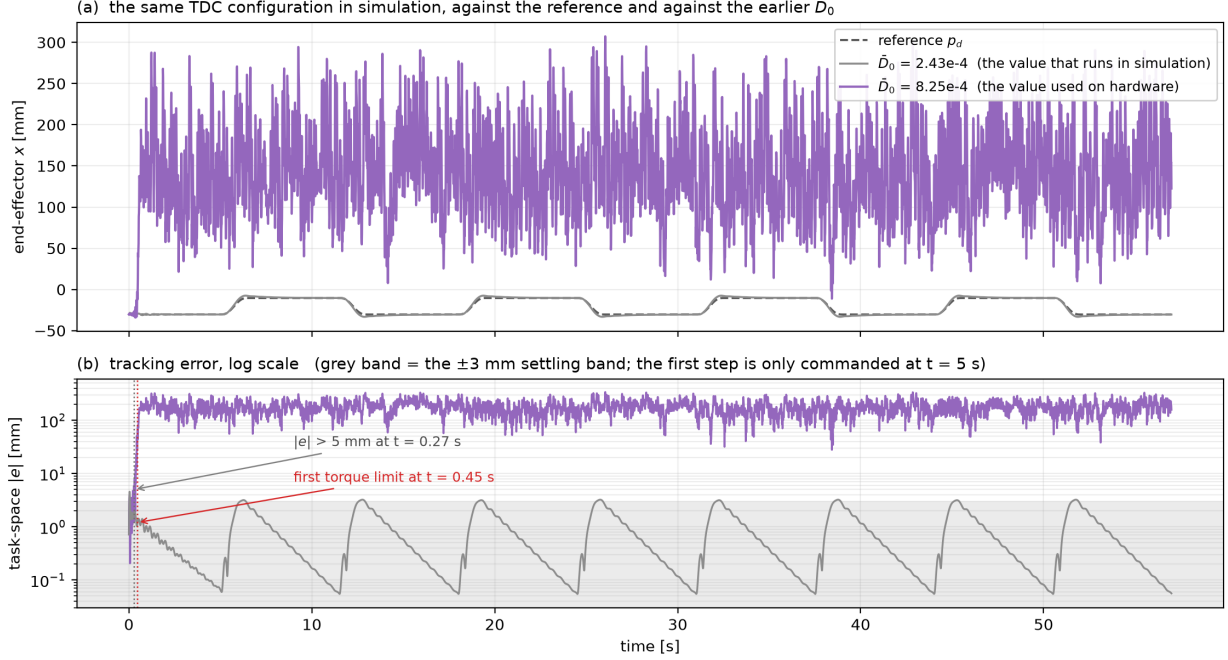


Figure 17: The hardware-verified TDC configuration, run in simulation. The earlier $\bar{D}_0$, which runs normally in simulation, is shown for reference.

The reason is the one the report has been building towards. $\bar{D}$ scales the loop gain, so raising $\bar{D}_0$ by $3.40\times$ raises joint 1's effective gains by $3.40\times$: $\bar{D}_0 K_p^{\text{TDC}}$ from 0.024 to 0.083 Nm/rad and $\bar{D}_0 K_d^{\text{TDC}}$ from 0.039 to 0.132 Nm/(rad/s). On hardware that additional authority is spent breaking stiction. The simulation has no Coulomb friction, so there is nothing there for it to break and the same setting is simply $3.4\times$ too much gain.

This is the sharpest instance of the caveat already stated in Section 5.1. For the PID's K_i^{PID} and the observer's $q_{\hat{d}}$ the simulation merely fails to show the benefit of a friction-fighting parameter. For $\bar{D}$ it does worse than that: it returns a confident negative verdict on the configuration that produces the best result in the entire study.

6.4 Repeatability

Three features of Figure 18 deserve to be named. First, the PID in band 2 flattens out at about 15 mm of the commanded 20 mm and only creeps to roughly 17 mm by $t = 6$ s; the whole bundle of 24 traces never reaches the setpoint within the dwell. Second, the MPC's spread in band 2 is visibly larger than in band 1, with per-run settled errors of 1.703, 1.460 and 0.951 mm. Third, the TDC bundles are the tightest of the six in both bands, which is the qualitative form of the overshoot standard deviations reported below.

Repeatability: all steps of all repeats, sign-normalised (dashed line = commanded 20 mm)

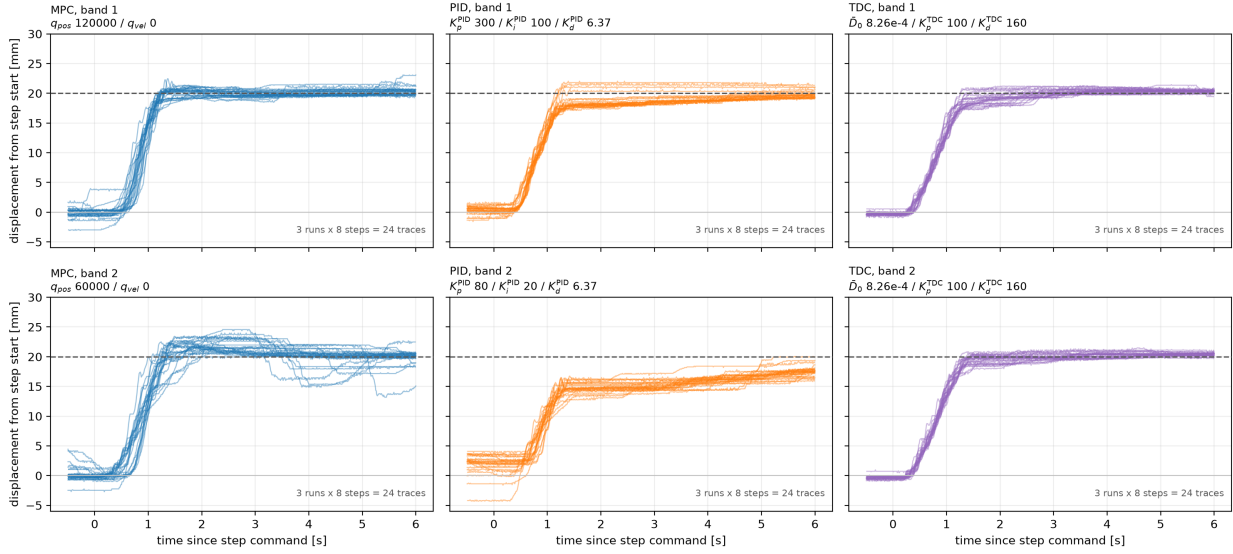


Figure 18: Every step of every repeat overlaid, sign-normalised so that outbound and return moves fall on top of each other. Three runs of eight steps in each of the six cells.

6.5 Main result: the three controllers

Table 8: Verified hardware performance, $n = 24$ per cell (3 runs $\times$ 8 steps). $|e_x|$ is the mean absolute error along the stepped axis over the last 2.0 s of each dwell. $|L|$ is not defined for TDC (Section 4.5).

Controller	Band	$ L $	$ e_x $ [mm]	rise [s]	overshoot [%]	settling [s]
MPC	1	0.255	0.504 ± 0.362	0.544 ± 0.155	$+5.68 \pm 3.50$	1.536
PID	1	0.261	0.663 ± 0.319	1.095 ± 0.471	$+0.28 \pm 3.58$	1.187
TDC	1	—	0.395 ± 0.209	0.985 ± 0.327	$+3.31 \pm 1.75$	1.185
MPC	2	0.156	0.864 ± 0.861	0.697 ± 0.312	$+11.31 \pm 5.27$	2.163
PID	2	0.195	2.823 ± 0.745	2.145 ± 2.849	-10.47 ± 4.73	5.415
TDC	2	—	0.428 ± 0.209	0.827 ± 0.094	$+3.95 \pm 1.49$	1.192

Welch tests on the settled $|e_x|$:

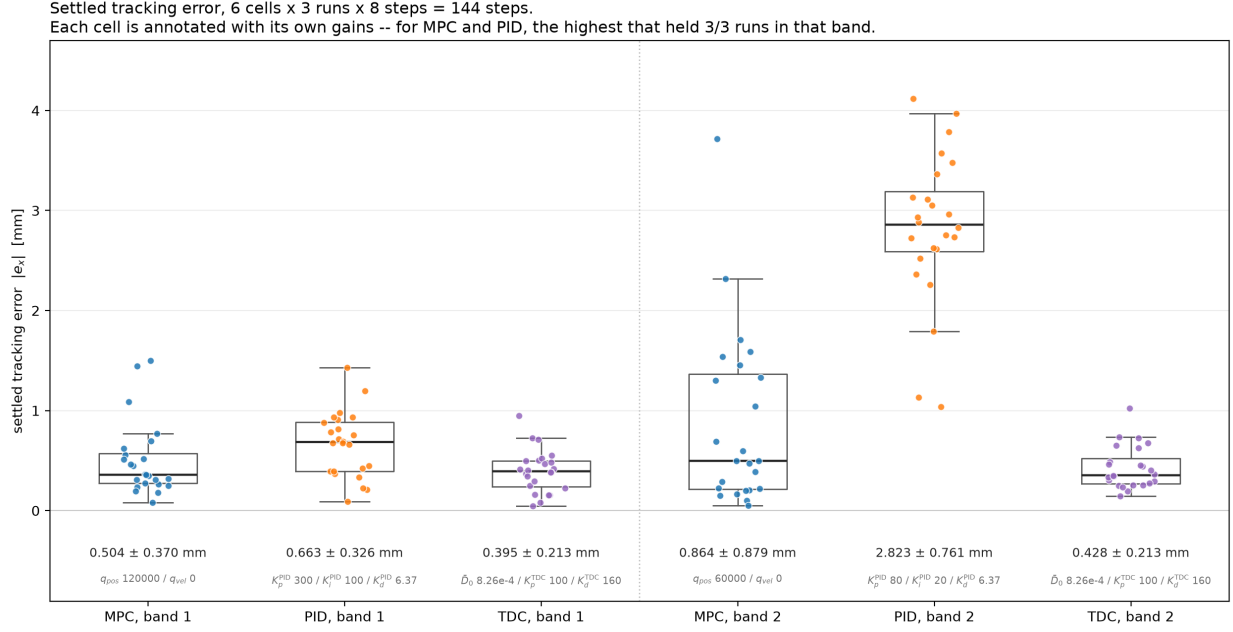


Figure 19: All 144 steps (3 controllers $\times$ 2 bands $\times$ 3 runs $\times$ 8 steps), with box plots of the settled $|e_x|$. Each cell is annotated with its own configuration.

Comparison	t	p	Verdict
Band 1: MPC vs PID	-1.59	$1.2 \cdot 10^{-1}$	not distinguishable
Band 1: TDC vs MPC	-1.25	$2.2 \cdot 10^{-1}$	not distinguishable
Band 1: TDC vs PID	-3.38	$1.7 \cdot 10^{-3}$	TDC better
Band 2: MPC vs PID	-8.25	$1.5 \cdot 10^{-10}$	MPC better
Band 2: TDC vs MPC	-2.36	$2.6 \cdot 10^{-2}$	TDC better
Band 2: TDC vs PID	-14.84	$2.2 \cdot 10^{-14}$	TDC better
MPC across bands	-1.85	$7.4 \cdot 10^{-2}$	not distinguishable
PID across bands	-12.78	$6.3 \cdot 10^{-14}$	significantly worse
TDC across bands	-0.54	$6.0 \cdot 10^{-1}$	not distinguishable

Reading the table. Four statements are supported by it.

1. **TDC has the lowest settled error at both postures** (0.395 and 0.428 mm). Against the PID this is significant in both bands; against the MPC it is significant in band 2 and not distinguishable in band 1.
2. **TDC is the only controller that is posture-insensitive on every metric at once.** Its settling time is unchanged between the bands (1.185 against 1.192 s), where the MPC goes from 1.536 to 2.163 s and the PID from 1.187 to 5.415 s. The MPC's settled error is also statistically unchanged across the bands, but its overshoot doubles and its settling time grows by 41 %.

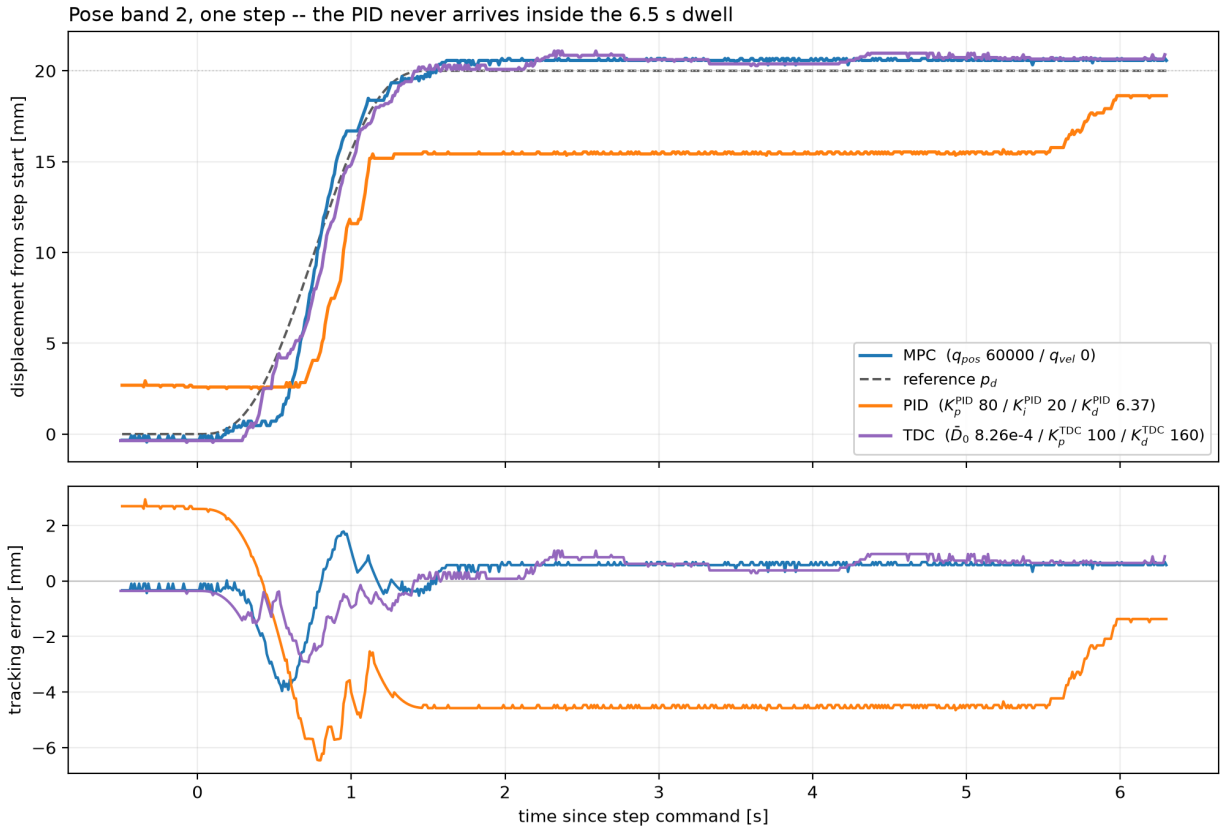


Figure 20: One step in posture band 2 in detail. The PID stops 4.6 mm short of the setpoint and holds there for five seconds, until the next step command; the MPC and TDC both arrive within 1.5 s.

3. **TDC is the most repeatable.** Its overshoot standard deviation is 1.75 and 1.49 % against the MPC’s 3.50 and 5.27 and the PID’s 3.58 and 4.73; its rise-time standard deviation in band 2 is 0.094 s against the PID’s 2.849 s.
4. **The MPC’s remaining advantage is speed.** It has the fastest rise time in both bands, 0.544 and 0.697 s against TDC’s 0.985 and 0.827 s. This is the one metric on which retuning $\bar{D}$ cost TDC something: the earlier $\bar{D}$ rose in 0.737 s.

Why the PID collapses in band 2 and the MPC does not. The explanation is in the composition of the commanded force, Eq. (20). Decomposing it over the runs, the rms of the observer term $-\mathbf{\Lambda}\hat{\mathbf{d}}$ relative to the proportional-derivative term $-\mathbf{K}[e; \dot{e}]$ is $15.9\times$ at $K_p^{\text{MPC}} = 2.2$, $4.8\times$ at $K_p^{\text{MPC}} = 40.0$ and $3.4\times$ at $K_p^{\text{MPC}} = 101.4$. The observer, acting as a slow disturbance integrator, supplies the majority of the steady-state authority at every gain tested, so cutting K_p^{MPC} removes only a minority of the MPC’s authority. The PID has no second channel: lowering its gain cuts K_p^{PID} and K_i^{PID} together and both of its routes to the setpoint shrink at once. In Figure 20 the consequence is visible directly — the PID reaches 15.4 mm of the commanded 20 mm and stops there, because at $K_p^{\text{PID}} = 80$ the proportional term can no longer break the remaining stiction and the integral term is clamped.

The difference between the bands is not the task-space inertia. The obvious candidate explanation for a posture-dependent result is that the arm presents a different apparent mass in the two bands, so $\mathbf{\Lambda}(\mathbf{q})$ was evaluated along the measured joint trajectories of both runs rather than at a single nominal pose:

	Λ_{xx} mean [kg]	range over the band [kg]	spread
band 1	0.6683	0.6542 – 0.6847	4.56 %
band 2	0.6513	0.6408 – 0.6661	3.88 %

The two band means differ by 2.55 % in Λ_{xx} and 4.36 % in Λ_{zz} — *less than the variation the arm already goes through within a single band* as it sweeps its own 20 mm interval (4.56 and 3.88 %). A quantity that changes by more within a band than between bands cannot be what separates them. The same holds for the conditioning of the Jacobian, measured the same way: $\text{cond}(J_{xz}) = 1.232$ (range 1.198–1.284) against 1.290 (1.230–1.336), two ranges that overlap. Whatever distinguishes the two postures is not kinematic or inertial; the remaining candidate, consistent with everything else in this chapter, is friction and the gravity preload that sets it.

6.6 Compute budget and achievable loop rate

The last measurement is a practical one, taken from the same logs.

Table 9: Per-tick computation cost including servo I/O, measured on the verified hardware runs. The control period is 10 ms.

Controller	mean [ms]	p99 [ms]	max [ms]	p99 as % of period
MPC (<code>a41_pos</code>)	8.17	10.49	13.64	105 %
Task-space PID (<code>pidC18_pos</code>)	6.73	8.87	37.57	89 %
TDC (<code>dbarbal.7</code>)	2.71	5.30	7.60	53 %

The MPC’s 99th-percentile tick cost **exceeds the control period**. It is compute-bound on this hardware: it cannot be run faster than 100 Hz here and this is a structural property of solving a QP every tick, not a tuning oversight. The same 8 ms solve time is one of the four terms in the $T_d \approx 30$ ms delay budget of Section 4.2, so the delay wall that limits the achievable gain and the compute budget that limits the achievable rate are two views of one quantity.

TDC costs roughly 5.5 ms per tick less than the MPC, the difference being the QP solve and is the only one of the three with substantial headroom on this hardware. Section 5.3 shows what that headroom is worth in simulation: overshoot falls from 14.3 to 1.0 % between 100 and 500 Hz; Section 6.7 shows what it is worth on hardware. The PID’s 6.73 ms is most likely communication-bound rather than computation-bound, since the control law itself is nearly free; that hypothesis has not been tested.

6.7 TDC at 500 Hz on hardware

Section 6.6 showed TDC alone has slack left in its per-tick budget; Section 5.3 showed what that slack is worth in simulation. This section closes the loop with the actual hardware run at 500 Hz, on the identical MPC-matched step reference of Table 8 and the identical repeatability protocol ($n = 24$, three runs of eight steps, both bands).

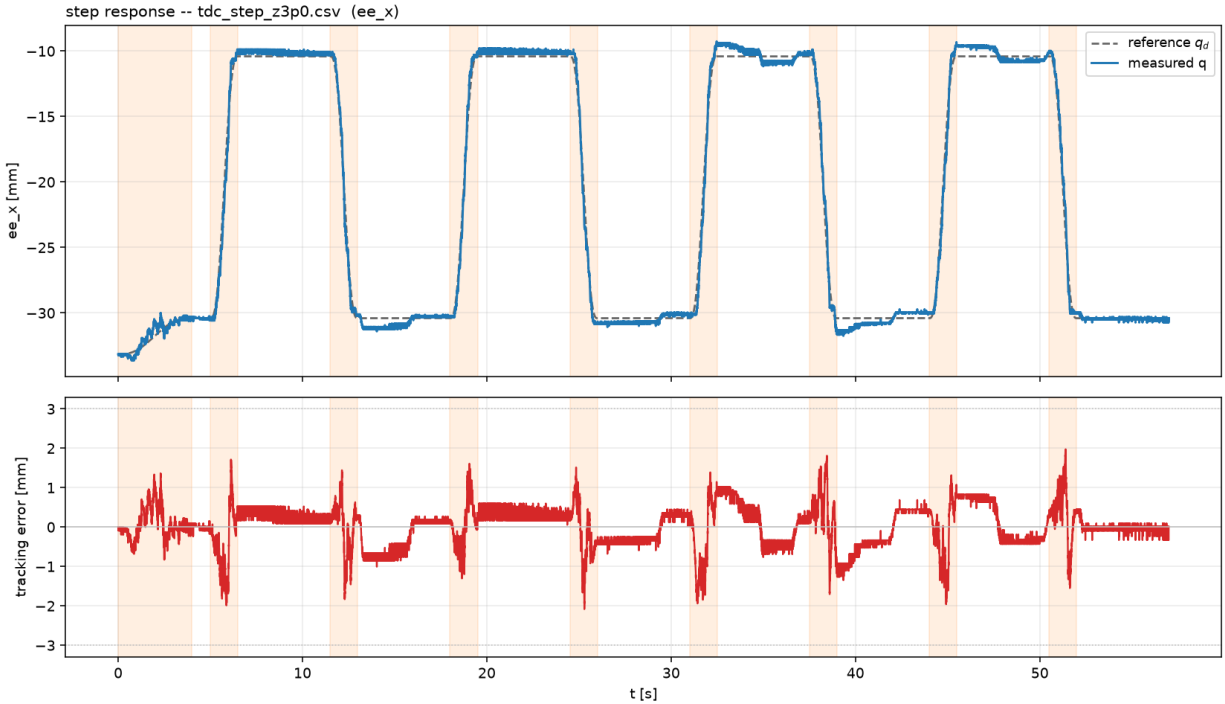


Figure 21: TDC at 500 Hz, band 1, one full run (`tdc_step_z3p0`). The first ~ 4 s (unshaded) is the homing ease-in settling into the reference’s own start pose and is excluded from every statistic below, the same handling Table 8 already gives step 0 elsewhere in this chapter.

Three of four metrics improve at 500 Hz on both bands: settled error falls 44 % (band 1) and 52 % (band 2), rise time falls 30 % and 20 %, settling time falls 11 % and 13 %. Overshoot does *not* follow – it is marginally higher at 500 Hz on both bands (+4.05 vs. +3.31 %, +4.22 vs. +3.95 %) – so the headroom Section 6.6 identified buys faster, tighter tracking, not a cleaner transient, and nothing here should be read as “500 Hz strictly dominates 100 Hz.”

Table 10: TDC on hardware, 100 Hz (repeated from Table 8) vs. 500 Hz, $n = 24$ per cell (3 runs $\times$ 8 steps).

Band	Rate	$ e_x $ [mm]	rise [s]	overshoot [%]	settling [s]
1	100 Hz	0.395 ± 0.209	0.985 ± 0.327	$+3.31 \pm 1.75$	1.185
1	500 Hz	0.222 ± 0.132	0.694 ± 0.052	$+4.05 \pm 1.64$	1.060 ± 0.038
2	100 Hz	0.428 ± 0.209	0.827 ± 0.094	$+3.95 \pm 1.49$	1.192
2	500 Hz	0.207 ± 0.123	0.663 ± 0.052	$+4.22 \pm 1.25$	1.039 ± 0.037

Against Table 6’s simulated 500 Hz prediction (RMSE 0.131 mm), the hardware settled error of 0.21–0.22 mm is worse by roughly 0.09 mm – the same qualitative gap Section 2.4 documents everywhere else in this report between an idealised model and a friction-dominated arm, not a contradiction of the simulation result.

What is, and is not, held fixed across the rate change. $K_p^{\text{TDC}} = 300$ and $\zeta = 3.0$ transfer unchanged from the 100 Hz configuration, but $\bar{D}$ does not: this run uses $\bar{D} = (2.4264, 2.7750, 0.3674) \cdot 10^{-4} \text{ kg m}^2$, chosen by a separate procedure (a $\text{dt}=0.002$ grid search in simulation, cross-validated against hold-from-rest at two postures, rather than the 12-point band-1 hardware sweep of Section 6.2 behind the report’s headline 100 Hz value $(8.26, 2.80, 1.52) \cdot 10^{-4}$). The two $\bar{D}$ values are not a “which is bigger” comparison – they are answers to the same tuning question asked at two different loop rates, each validated on its own terms, not against each other.

6.8 Native Position Control Mode

The sibling **Baseline** project commands no torque. It streams a desired joint angle (`Goal_Position`) at 500 Hz through the servo firmware’s own position loop, with no task-space control law, no disturbance observer and no gravity or friction compensation. It runs the identical step reference (`Baseline/references/step_x_ref.csv` is a byte-for-byte copy of this project’s own reference file) under the identical repeatability protocol.

The servo ships with $I = 0$ (pure proportional control), which leaves a gravity-dependent steady-state error nothing integrates away – the factory row of Table 11. Table 11 and Figure 23 show the tuning that followed it, chronologically: eleven probes, the Position D/I/P gains (joints 0–2) swept jointly, each scored by step-period RMSE the same way this whole chapter scores everything.

Position Control Mode, once tuned, settles tighter and cleaner than TDC at the same rate on both bands – roughly $2\times$ lower settled error and essentially no overshoot ($+0.5\%$ against TDC’s $+4\text{--}4.2\%$) – but is slower to get there, by 60–100 ms on both rise and settling time. The tuned gains, $D = 100, 50, 100$, $I = 60, 70, 60$, $P = 1800, 800, 1500$ (joints 0–2), were reached by eleven probes (D, I, P swept jointly, scored by step-period RMSE the same way this whole chapter scores everything), moving only what Dynamixel’s own control table exposes – never a code change on either side of the interface – and took the steady-state offset from the factory value above, 0.955 mm, down to 0.079 mm overall.

A position loop streaming `Goal_Position` has no commandable task-space stiffness; Section 7 takes up what that means for the comparison.

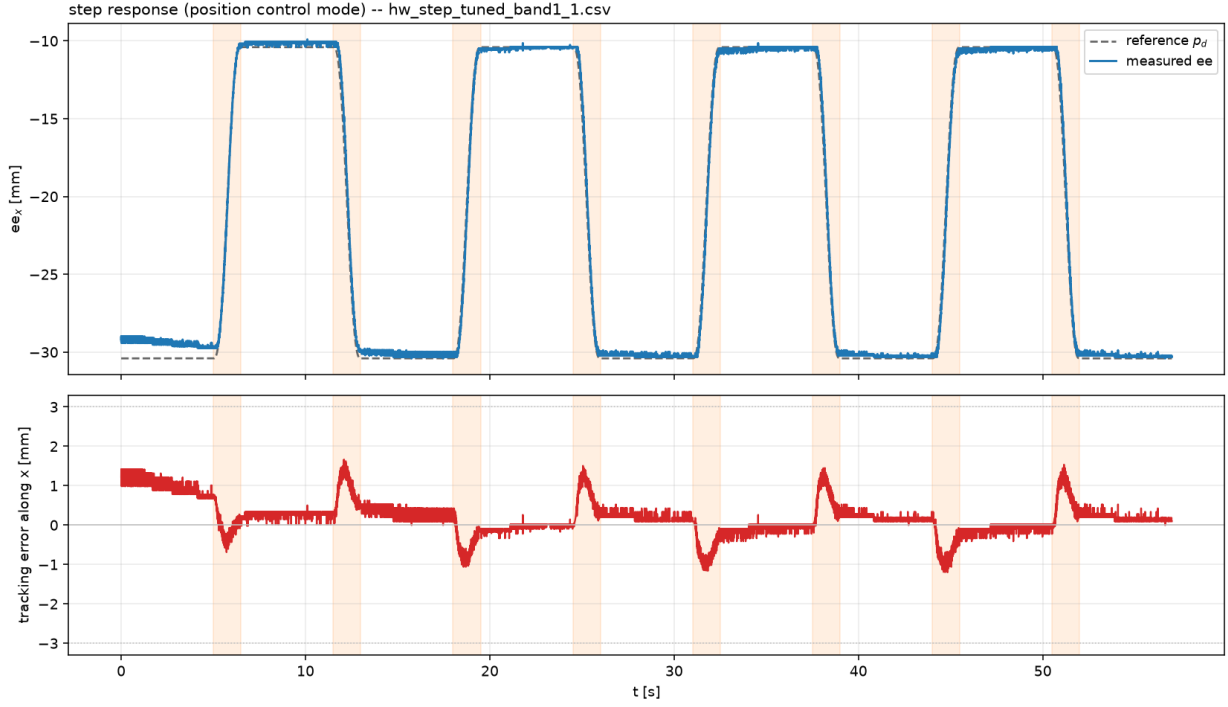


Figure 22: Native Position Control Mode, band 1, tuned gains, one run of three (hw_step_tuned_band1_1).

Table 11: Native Position Control Mode's tuning sweep, in the order it was run. Probe 11 (bold) is the configuration used everywhere else in this section.

Probe	D	I	P	RMSE, 2-D [mm]
Factory	0,0,0	0,0,0	800,800,800	1.457
1	1,0,1	50,30,40	800,800,800	0.830
2	3,0,1	60,30,50	800,800,800	0.809
3	5,0,1	60,50,60	800,800,800	0.776
4	8,0,1	60,60,70	800,800,800	0.763
5	10,1,1	60,70,80	800,800,800	0.769
6	20,1,1	60,70,80	1000,800,800	0.628
7	40,1,1	60,70,80	1500,800,800	0.544
8	50,1,25	60,70,80	1500,800,800	0.535
9	100,50,50	60,70,80	1500,800,800	0.539
10	100,50,100	60,70,60	1500,800,800	0.540
11	100,50,100	60,70,60	1800,800,1500	0.470

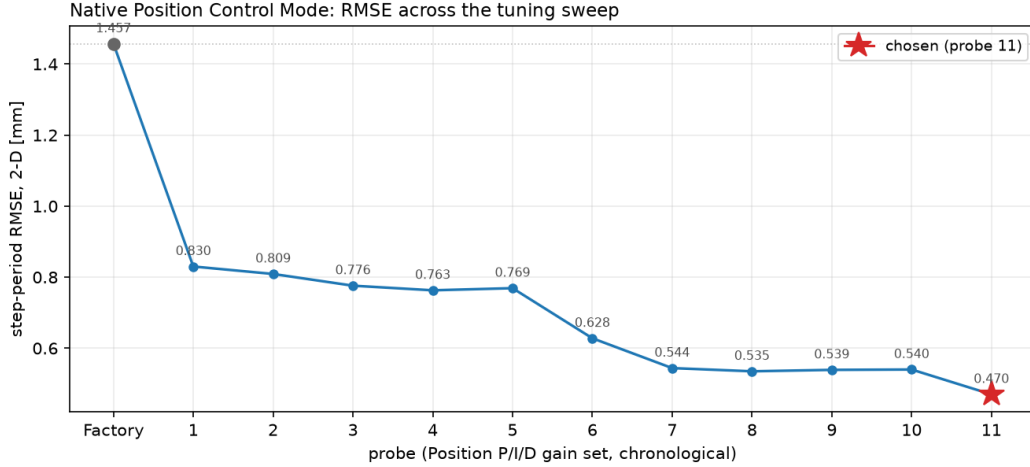


Figure 23: RMSE across the sweep in Table 11. Mostly monotone, with two small upticks (probes 5 and 9–10) that stay well inside the run-to-run noise seen elsewhere in this chapter; both places P actually moved (probes 6–8, then 11) are exactly where RMSE actually drops.

Table 12: Native Position Control Mode (tuned gains, probe 11 of Table 11) vs. TDC at 500 Hz (Table 10), $n = 24$ per cell (3 runs $\times$ 8 steps), both on the identical step reference.

Band	Controller	$ e_x $ [mm]	rise [s]	overshoot [%]	settling [s]
1	Position Control Mode	0.107 ± 0.078	0.746 ± 0.008	$+0.52 \pm 0.93$	1.102 ± 0.015
1	TDC (500 Hz)	0.222 ± 0.132	0.694 ± 0.052	$+4.05 \pm 1.64$	1.060 ± 0.038
2	Position Control Mode	0.050 ± 0.010	0.760 ± 0.011	$+0.50 \pm 0.25$	1.108 ± 0.005
2	TDC (500 Hz)	0.207 ± 0.123	0.663 ± 0.052	$+4.22 \pm 1.25$	1.039 ± 0.037

7 Synthesis and Limitations

What these experiments support. The arm is friction-dominated, and this can be stated without reference to any controller. The measured task-space breakaway force is 0.94 N. The rigid-body model matches the arm statically to within the encoder noise floor, at worst 0.031 degrees on any joint, yet replayed open-loop it diverges on all three joints within 0.1 s, worst on joint 3, until one constant is added to the inertia diagonal, which reduces the 0.1 s prediction RMSE of joint 3 from 137.0 degrees to 2.87 degrees. The open-loop test establishes that this constant is needed, not what its value should be. The one place the reading is tested rather than inferred is a negative result: the TDC configuration that produced the best result of the whole study on hardware diverges within 0.45 s in a frictionless simulation.

What these experiments do not support. The main comparison is between two task-space controllers at 100 Hz. Two supplementary configurations, TDC at 500 Hz and the servo’s native Position Control Mode, settle tighter than both and differ from them in more than one respect at once: both run at five times the rate and both track a joint-space reference, and the position loop additionally closes inside the servo firmware, so the delay budget of roughly 30 ms established in Section 4.2 does not apply to it. TDC nevertheless still commands torque; what it shares with the position loop is the reference space, not the actuation level. Neither of the two properties that motivate task-space control was exercised here. Commandable compliance was not: no external force was applied in any hardware run, and the position loop has no commandable task-space stiffness to begin with. Redundancy resolution was not either, since no joint was ever obstructed. The metric used throughout, step-tracking accuracy at a fixed posture with no contact, is one on which a joint-space controller has no structural disadvantage, so these results say nothing about which formulation is preferable in general. Four further limits stand: the two task-space controllers had to be detuned for band 2, so their band-2 numbers confound posture with gain; each point of the $\bar{D}$ sweep is a single run except the chosen one; the observer’s estimate is never compared against a measured disturbance; and the PID’s force clamp is 3.0 N while the MPC’s box is 300 N against a peak commanded force of 1.41 N, so the two are not constrained alike.